Is the zebra to the left of the bench black or white?



Gold: white
Historical All-Writer: White
Historical Semantic-k: Black

P-1 is separate (not in R ranking):
L8-11: M=0.0398
L0-15: M=0.1133
L16-19: M=0.0703
L0-31: M=0.0866

Question groups below: L0-15 averages over all heads/layers/subwords. M=Mass, E=Excess, En=mean per-head Enrichment. Green=historical selected; gray=omitted; orange border=relative omitted top5, not a causal threshold.

Is @583 n=1 M 0.1996 E -0.7867 En 0.0268	the @584 n=1 M 0.2129 E -0.7717 En 0.0253	zebra @585 n=3 M 0.2467 E -0.7345 En 0.0366	to @588 n=1 M 0.1821 E -0.7958 En 0.0268	the @589 n=1 M 0.1879 E -0.7883 En 0.0285	left @590 n=1 M 0.1648 E -0.8098 En 0.0267
of @591 n=1 M 0.1492 E -0.8237 En 0.0247	the @592 n=1 M 0.1837 E -0.7877 En 0.0275	bench @593 n=2 M 0.2089 E -0.7600 En 0.0321	black @595 n=1 M 0.1683 E -0.7982 En 0.0301	or @596 n=1 M 0.1279 E -0.8370 En 0.0245	white @597 n=1 M 0.1560 E -0.8073 En 0.0283
? @598 n=1 M 0.1187 E -0.8429 En 0.0216					

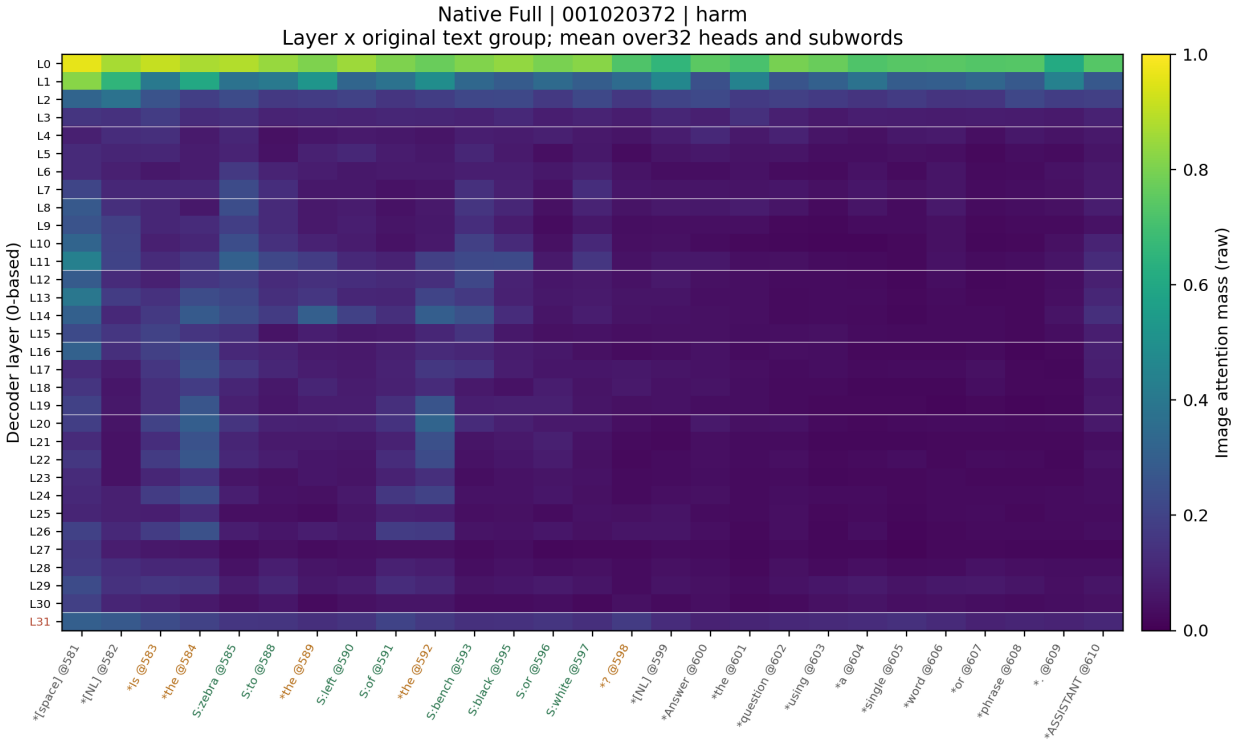
Other R groups (Instruction and Assistant/template); all historically omitted. Exact subwords and all-layer metrics are preserved in CSV.

Group / positions	Scope	Mass	Excess	Enrich.
[space] @581	Assistant/template	0.3311	-0.6586	0.0614
[NL] @582	Assistant/template	0.2355	-0.7525	0.0252
[NL] @599	Assistant/template	0.1312	-0.8288	0.0231
Answer @600	Instruction	0.1220	-0.8364	0.0211
the @601	Instruction	0.1271	-0.8297	0.0237
question @602	Instruction	0.1160	-0.8392	0.0264
using @603	Instruction	0.1081	-0.8455	0.0252
a @604	Instruction	0.1119	-0.8401	0.0241
single @605	Instruction	0.1056	-0.8449	0.0248
word @606	Instruction	0.1145	-0.8344	0.0255
or @607	Instruction	0.1029	-0.8445	0.0242
phrase @608	Instruction	0.1068	-0.8390	0.0246
. @609	Instruction	0.1143	-0.8299	0.0235
ASSISTANT @610,611,612,613	Assistant/template	0.1452	-0.7952	0.0333

Template [space] immediately after Image is distinct from the ASSISTANT word. Negative Excess and Enrichment below1 mean the observed mass is below the uniform per-key baseline.

01 | 001020372 | ChooseAttr | harm

NATIVE FULL OBSERVATION | layer heatmap and omitted Top5 | no new answer generation



S: historical Semantic-selected; * omitted. Orange labels: omitted Question top5 in L0-15. P-1 excluded. L31 is observational only.

Omitted whole-group Top5 by scope (all32 heads averaged, no single-head maximum ranking):

L8-11

Question: the @584 0.114; Is @583 0.106; the @589 0.103; the @592 0.098; ? @598 0.045

Instruction: word @606 0.052; Answer @600 0.051; the @601 0.045; . @609 0.044; question @602 0.037

Assistant/template: [space] @581 0.323; [NL] @582 0.181; ASSISTANT @610 0.088; [NL] @599 0.055

L0-15

Question: the @584 0.213; Is @583 0.200; the @589 0.188; the @592 0.184; ? @598 0.119

Instruction: the @601 0.127; Answer @600 0.122; question @602 0.116; word @606 0.114; . @609 0.114

Assistant/template: [space] @581 0.331; [NL] @582 0.236; ASSISTANT @610 0.145; [NL] @599 0.131

L16-19

Question: the @584 0.229; the @592 0.167; Is @583 0.154; the @589 0.083; ? @598 0.053

Instruction: Answer @600 0.046; the @601 0.037; or @607 0.034; question @602 0.034; using @603 0.032

Assistant/template: [space] @581 0.195; [NL] @582 0.084; ASSISTANT @610 0.075; [NL] @599 0.047

L31 R outputs cannot propagate onward to the first-answer P-1 position in the answer-image-blocked causal graph. High L31 mass is not evidence of functional necessity. All answers shown are historical, not Native Full generation in this run.

Does the bathtub look black or cream colored?



Gold: cream colored
Historical All-Writer: Cream colored
Historical Semantic-k: Black

P-1 is separate (not in R ranking):
L8-11: M=0.0605
L0-15: M=0.1281
L16-19: M=0.1101
L0-31: M=0.1119

Question groups below: L0-15 averages over all heads/layers/subwords. M=Mass, E=Excess, En=mean per-head Enrichment. Green=historical selected; gray=omitted; orange border=relative omitted top5, not a causal threshold.

Does	the	bathtub	look	black	or
@583 n=1	@584 n=1	@585 n=3	@588 n=1	@589 n=1	@590 n=1
M 0.2102 E -0.7761	M 0.2230 E -0.7616	M 0.2424 E -0.7388	M 0.1932 E -0.7848	M 0.2078 E -0.7685	M 0.1453 E -0.8293
En 0.0293	En 0.0257	En 0.0369	En 0.0290	En 0.0326	En 0.0255

cream	colored	?
@591 n=2	@593 n=1	@594 n=1
M 0.1722 E -0.8000	M 0.1660 E -0.8037	M 0.1387 E -0.8294
En 0.0308	En 0.0333	En 0.0247

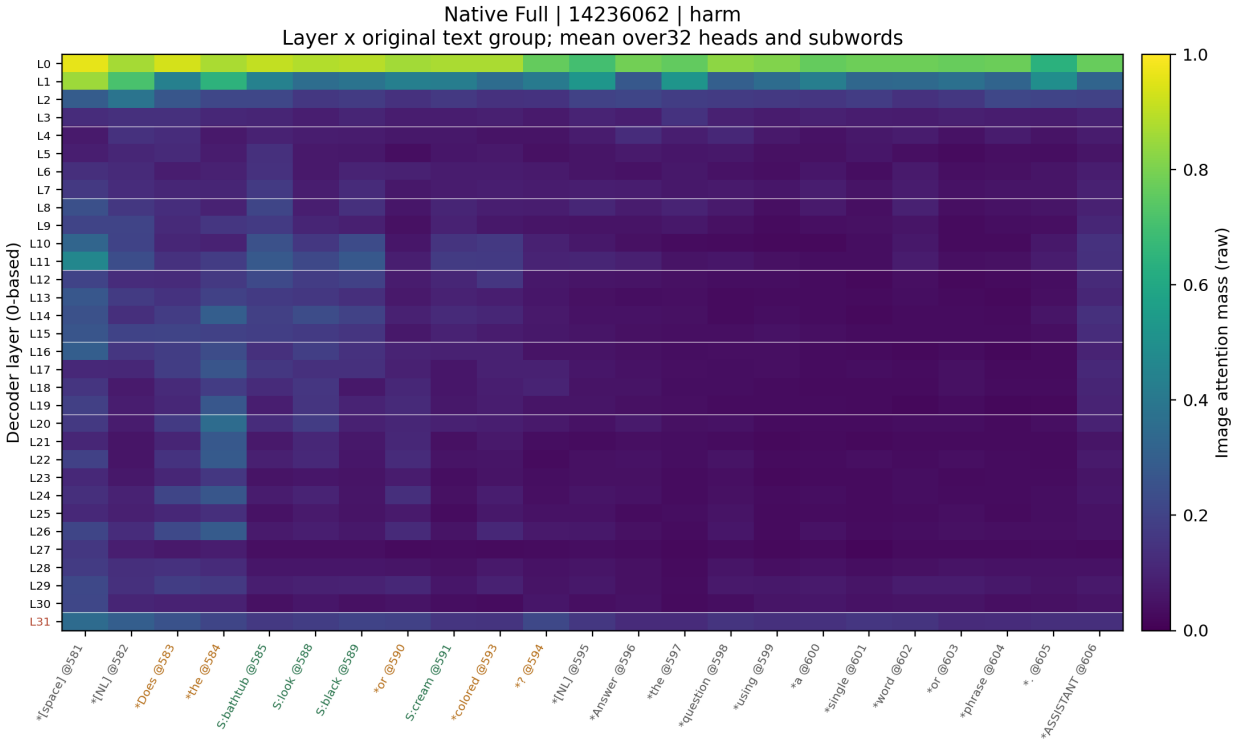
Other R groups (Instruction and Assistant/template); all historically omitted. Exact subwords and all-layer metrics are preserved in CSV.

Group / positions	Scope	Mass	Excess	Enrich.
[space] @581	Assistant/template	0.3071	-0.6826	0.0523
[NL] @582	Assistant/template	0.2526	-0.7354	0.0265
[NL] @595	Assistant/template	0.1458	-0.8206	0.0244
Answer @596	Instruction	0.1330	-0.8319	0.0230
the @597	Instruction	0.1435	-0.8197	0.0263
question @598	Instruction	0.1274	-0.8342	0.0286
using @599	Instruction	0.1194	-0.8406	0.0273
a @600	Instruction	0.1246	-0.8338	0.0256
single @601	Instruction	0.1158	-0.8410	0.0263
word @602	Instruction	0.1274	-0.8278	0.0273
or @603	Instruction	0.1136	-0.8401	0.0256
phrase @604	Instruction	0.1173	-0.8348	0.0268
. @605	Instruction	0.1273	-0.8232	0.0256
ASSISTANT @606,607,608,609	Assistant/template	0.1666	-0.7800	0.0399

Template [space] immediately after Image is distinct from the ASSISTANT word. Negative Excess and Enrichment below1 mean the observed mass is below the uniform per-key baseline.

02 | 14236062 | ChooseAttr | harm

NATIVE FULL OBSERVATION | layer heatmap and omitted Top5 | no new answer generation



S: historical Semantic-selected; * omitted. Orange labels: omitted Question top5 in L0-15. P-1 excluded. L31 is observational only.

Omitted whole-group Top5 by scope (all32 heads averaged, no single-head maximum ranking):

L8-11

Question: the @584 0.132; Does @583 0.126; colored @593 0.121; ? @594 0.081; or @590 0.062

Instruction: word @602 0.072; Answer @596 0.065; the @597 0.062; . @605 0.060; question @598 0.051

Assistant/template: [space] @581 0.310; [NL] @582 0.202; ASSISTANT @606 0.121; [NL] @595 0.082

L0-15

Question: the @584 0.223; Does @583 0.210; colored @593 0.166; or @590 0.145; ? @594 0.139

Instruction: the @597 0.144; Answer @596 0.133; word @602 0.127; question @598 0.127; . @605 0.127

Assistant/template: [space] @581 0.307; [NL] @582 0.253; ASSISTANT @606 0.167; [NL] @595 0.146

L16-19

Question: the @584 0.236; Does @583 0.149; or @590 0.105; colored @593 0.086; ? @594 0.077

Instruction: Answer @596 0.049; or @603 0.039; the @597 0.039; question @598 0.037; using @599 0.036

Assistant/template: [space] @581 0.190; [NL] @582 0.106; ASSISTANT @606 0.106; [NL] @595 0.056

L31 R outputs cannot propagate onward to the first-answer P-1 position in the answer-image-blocked causal graph. High L31 mass is not evidence of functional necessity. All answers shown are historical, not Native Full generation in this run.

Are these penguins or panda bears?



Gold: panda bears
Historical All-Writer: Panda bears
Historical Semantic-k: Panda

P-1 is separate (not in R ranking):
L8-11: M=0.0925
L0-15: M=0.1463
L16-19: M=0.1151
L0-31: M=0.1185

Question groups below: L0-15 averages over all heads/layers/subwords. M=Mass, E=Excess, En=mean per-head Enrichment. Green=historical selected; gray=omitted; orange border=relative omitted top5, not a causal threshold.

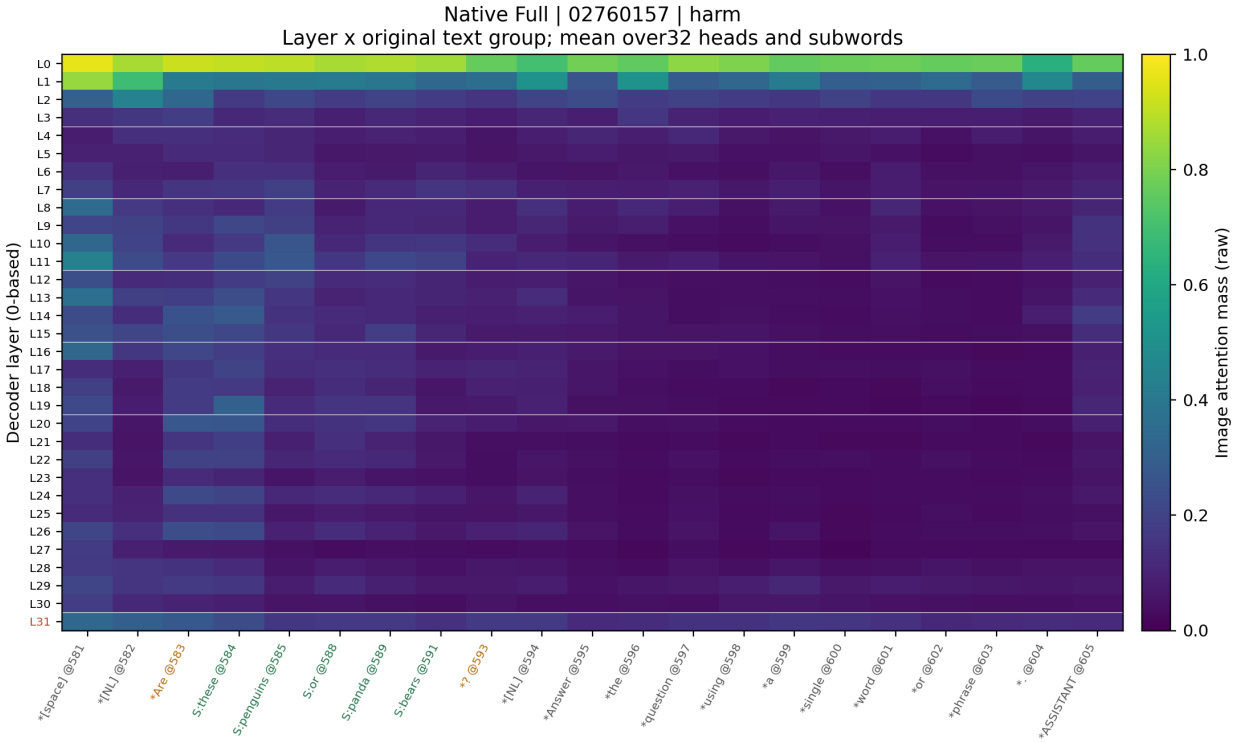
Are	these	penguins	or	panda	bears
@583 n=1	@584 n=1	@585 n=3	@588 n=1	@589 n=2	@591 n=2
M 0.2333 E -0.7530	M 0.2353 E -0.7493	M 0.2342 E -0.7470	M 0.1731 E -0.8048	M 0.1922 E -0.7832	M 0.1792 E -0.7929
En 0.0273	En 0.0268	En 0.0327	En 0.0262	En 0.0310	En 0.0307

?
@593 n=1
M 0.1495 E -0.8202
En 0.0238

Other R groups (Instruction and Assistant/template); all historically omitted. Exact subwords and all-layer metrics are preserved in CSV.

Group / positions	Scope	Mass	Excess	Enrich.
[space] @581	Assistant/template	0.3229	-0.6668	0.0605
[NL] @582	Assistant/template	0.2525	-0.7355	0.0274
[NL] @594	Assistant/template	0.1642	-0.8038	0.0259
Answer @595	Instruction	0.1417	-0.8247	0.0226
the @596	Instruction	0.1524	-0.8124	0.0258
question @597	Instruction	0.1353	-0.8279	0.0283
using @598	Instruction	0.1226	-0.8390	0.0268
a @599	Instruction	0.1284	-0.8316	0.0250
single @600	Instruction	0.1196	-0.8388	0.0256
word @601	Instruction	0.1344	-0.8224	0.0267
or @602	Instruction	0.1157	-0.8396	0.0246
phrase @603	Instruction	0.1201	-0.8336	0.0260
. @604	Instruction	0.1318	-0.8203	0.0247
ASSISTANT @605,606,607,608	Assistant/template	0.1693	-0.7788	0.0364

Template [space] immediately after Image is distinct from the ASSISTANT word. Negative Excess and Enrichment below1 mean the observed mass is below the uniform per-key baseline.



S: historical Semantic-selected; * omitted. Orange labels: omitted Question top5 in L0-15. P-1 excluded. L31 is observational only.

Omitted whole-group Top5 by scope (all32 heads averaged, no single-head maximum ranking):

L8-11

Question: Are @583 0.146; ? @593 0.097

Instruction: word @601 0.087; the @596 0.079; Answer @595 0.078; . @604 0.071; question @597 0.064

Assistant/template: [space] @581 0.332; [NL] @582 0.198; ASSISTANT @605 0.131; [NL] @594 0.109

L0-15

Question: Are @583 0.233; ? @593 0.149

Instruction: the @596 0.152; Answer @595 0.142; question @597 0.135; word @601 0.134; . @604 0.132

Assistant/template: [space] @581 0.323; [NL] @582 0.252; ASSISTANT @605 0.169; [NL] @594 0.164

L16-19

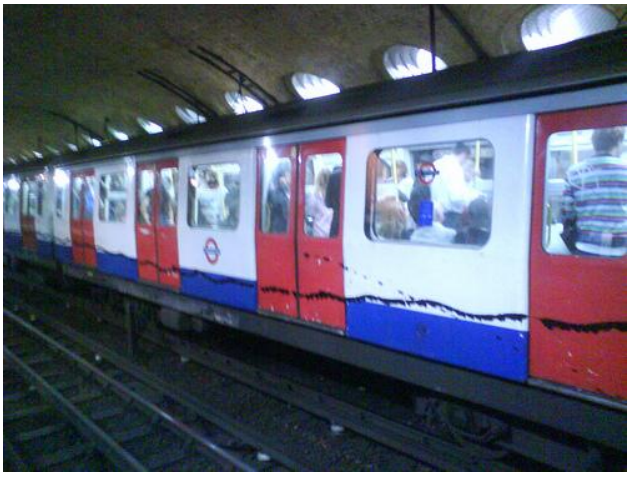
Question: Are @583 0.182; ? @593 0.086

Instruction: Answer @595 0.060; the @596 0.046; question @597 0.040; using @598 0.040; or @602 0.038

Assistant/template: [space] @581 0.217; [NL] @582 0.100; ASSISTANT @605 0.097; [NL] @594 0.092

L31 R outputs cannot propagate onward to the first-answer P-1 position in the answer-image-blocked causal graph. High L31 mass is not evidence of functional necessity. All answers shown are historical, not Native Full generation in this run.

Is this a truck or a train?



Gold: train
Historical All-Writer: Train
Historical Semantic-k: Truck

P-1 is separate (not in R ranking):
L8-11: M=0.0886
L0-15: M=0.1455
L16-19: M=0.1173
L0-31: M=0.1135

Question groups below: L0-15 averages over all heads/layers/subwords. M=Mass, E=Excess, En=mean per-head Enrichment. Green=historical selected; gray=omitted; orange border=relative omitted top5, not a causal threshold.

Is	this	a	truck	or	a
@583 n=1	@584 n=1	@585 n=1	@586 n=2	@588 n=1	@589 n=1
M 0.2078 E -0.7785	M 0.2037 E -0.7809	M 0.2180 E -0.7650	M 0.2206 E -0.7598	M 0.1602 E -0.8178	M 0.1696 E -0.8066
En 0.0263	En 0.0265	En 0.0274	En 0.0333	En 0.0263	En 0.0261

train	?
@590 n=1	@591 n=1
M 0.1953 E -0.7793	M 0.1552 E -0.8178
En 0.0312	En 0.0229

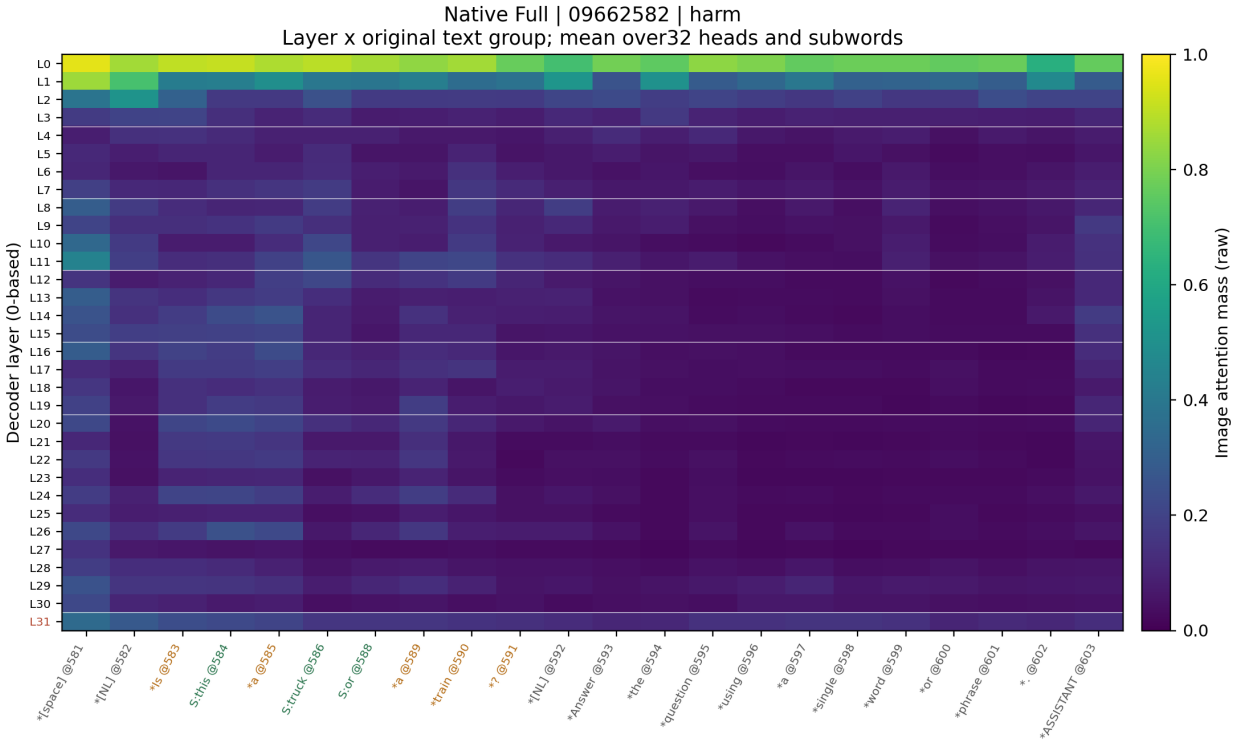
Other R groups (Instruction and Assistant/template); all historically omitted. Exact subwords and all-layer metrics are preserved in CSV.

Group / positions	Scope	Mass	Excess	Enrich.
[space] @581	Assistant/template	0.3175	-0.6722	0.0569
[NL] @582	Assistant/template	0.2437	-0.7443	0.0269
[NL] @592	Assistant/template	0.1652	-0.8061	0.0240
Answer @593	Instruction	0.1368	-0.8329	0.0207
the @594	Instruction	0.1457	-0.8224	0.0243
question @595	Instruction	0.1305	-0.8359	0.0268
using @596	Instruction	0.1192	-0.8457	0.0253
a @597	Instruction	0.1227	-0.8405	0.0240
single @598	Instruction	0.1166	-0.8450	0.0241
word @599	Instruction	0.1321	-0.8279	0.0249
or @600	Instruction	0.1112	-0.8472	0.0231
phrase @601	Instruction	0.1181	-0.8388	0.0238
. @602	Instruction	0.1304	-0.8249	0.0233
ASSISTANT @603,604,605,606	Assistant/template	0.1730	-0.7783	0.0384

Template [space] immediately after Image is distinct from the ASSISTANT word. Negative Excess and Enrichment below1 mean the observed mass is below the uniform per-key baseline.

04 | 09662582 | ChooseCat | harm

NATIVE FULL OBSERVATION | layer heatmap and omitted Top5 | no new answer generation



S: historical Semantic-selected; * omitted. Orange labels: omitted Question top5 in L0-15. P-1 excluded. L31 is observational only.

Omitted whole-group Top5 by scope (all32 heads averaged, no single-head maximum ranking):

L8-11

Question: train @590 0.178; a @585 0.148; Is @583 0.117; a @589 0.115; ? @591 0.109

Instruction: word @599 0.084; Answer @593 0.073; . @602 0.069; the @594 0.068; question @595 0.055

Assistant/template: [space] @581 0.321; [NL] @582 0.166; ASSISTANT @603 0.138; [NL] @592 0.123

L0-15

Question: a @585 0.218; Is @583 0.208; train @590 0.195; a @589 0.170; ? @591 0.155

Instruction: the @594 0.146; Answer @593 0.137; word @599 0.132; question @595 0.131; . @602 0.130

Assistant/template: [space] @581 0.318; [NL] @582 0.244; ASSISTANT @603 0.173; [NL] @592 0.165

L16-19

Question: a @585 0.180; Is @583 0.161; a @589 0.137; train @590 0.101; ? @591 0.071

Instruction: Answer @593 0.053; the @594 0.040; question @595 0.039; or @600 0.037; using @596 0.037

Assistant/template: [space] @581 0.191; ASSISTANT @603 0.102; [NL] @582 0.094; [NL] @592 0.074

L31 R outputs cannot propagate onward to the first-answer P-1 position in the answer-image-blocked causal graph. High L31 mass is not evidence of functional necessity. All answers shown are historical, not Native Full generation in this run.

Is the man to the left or to the right of the boy that is wearing a coat?



Gold: right
Historical All-Writer: Right
Historical Semantic-k: Left

P-1 is separate (not in R ranking):
L8-11: M=0.0583
L0-15: M=0.1364
L16-19: M=0.1163
L0-31: M=0.1084

Question groups below: L0-15 averages over all heads/layers/subwords. M=Mass, E=Excess, En=mean per-head Enrichment. Green=historical selected; gray=omitted; orange border=relative omitted top5, not a causal threshold.

Is @583 n=1 M 0.2296 E -0.7567 En 0.0272	the @584 n=1 M 0.2515 E -0.7331 En 0.0266	man @585 n=1 M 0.2988 E -0.6841 En 0.0358	to @586 n=1 M 0.2168 E -0.7645 En 0.0275	the @587 n=1 M 0.2294 E -0.7502 En 0.0281	left @588 n=1 M 0.1934 E -0.7845 En 0.0274
or @589 n=1 M 0.1539 E -0.8223 En 0.0241	to @590 n=1 M 0.1588 E -0.8159 En 0.0247	the @591 n=1 M 0.1758 E -0.7972 En 0.0245	right @592 n=1 M 0.1692 E -0.8022 En 0.0262	of @593 n=1 M 0.1697 E -0.8000 En 0.0262	the @594 n=1 M 0.2026 E -0.7655 En 0.0288
boy @595 n=1 M 0.2460 E -0.7204 En 0.0364	that @596 n=1 M 0.1854 E -0.7795 En 0.0276	is @597 n=1 M 0.1884 E -0.7748 En 0.0276	wearing @598 n=2 M 0.2265 E -0.7343 En 0.0377	a @600 n=1 M 0.2095 E -0.7489 En 0.0338	coat @601 n=1 M 0.2242 E -0.7326 En 0.0384
? @602 n=1 M 0.1635 E -0.7917 En 0.0262					

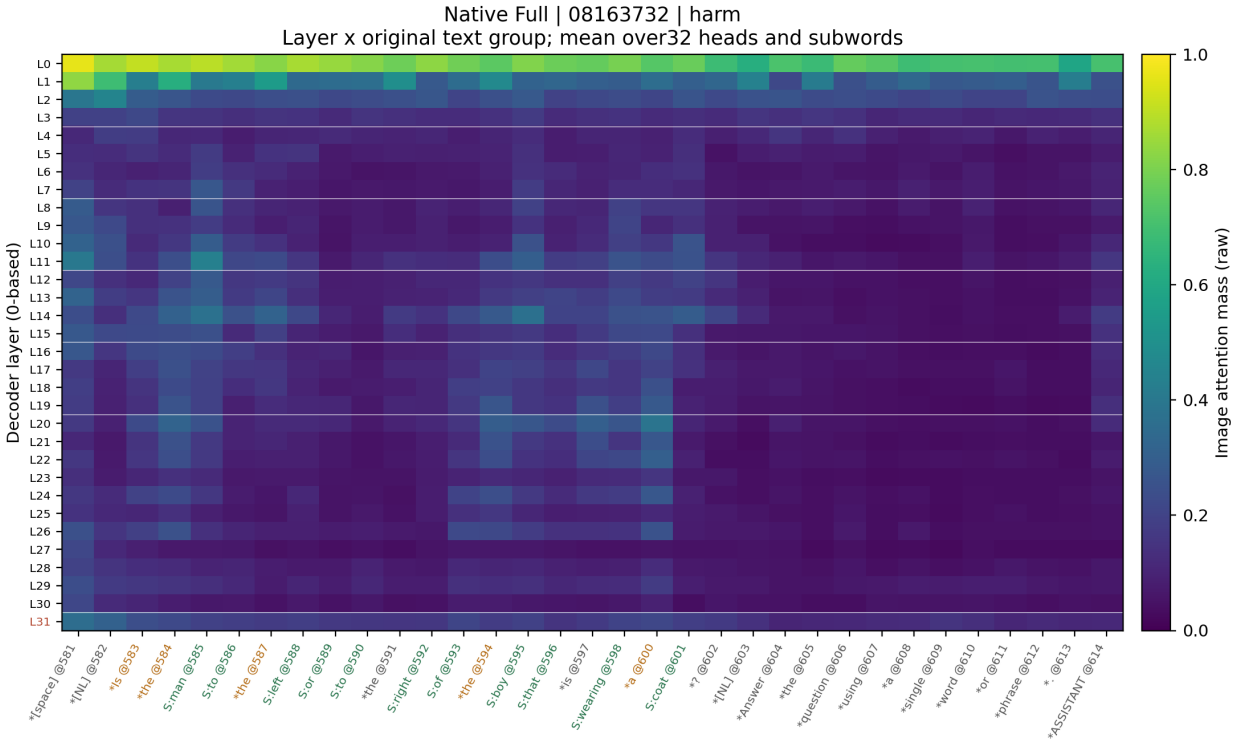
Other R groups (Instruction and Assistant/template); all historically omitted. Exact subwords and all-layer metrics are preserved in CSV.

Group / positions	Scope	Mass	Excess	Enrich.
[space] @581	Assistant/template	0.3303	-0.6594	0.0568
[NL] @582	Assistant/template	0.2673	-0.7207	0.0282
[NL] @603	Assistant/template	0.1546	-0.7990	0.0262
Answer @604	Instruction	0.1379	-0.8142	0.0223
the @605	Instruction	0.1433	-0.8072	0.0255
question @606	Instruction	0.1355	-0.8135	0.0276
using @607	Instruction	0.1255	-0.8218	0.0260
a @608	Instruction	0.1273	-0.8185	0.0252
single @609	Instruction	0.1226	-0.8217	0.0251
word @610	Instruction	0.1337	-0.8090	0.0262
or @611	Instruction	0.1164	-0.8247	0.0250
phrase @612	Instruction	0.1225	-0.8172	0.0256
. @613	Instruction	0.1310	-0.8071	0.0256
ASSISTANT @614,615,616,617	Assistant/template	0.1660	-0.7683	0.0518

Template [space] immediately after Image is distinct from the ASSISTANT word. Negative Excess and Enrichment below1 mean the observed mass is below the uniform per-key baseline.

05 | 08163732 | ChooseRel | harm

NATIVE FULL OBSERVATION | layer heatmap and omitted Top5 | no new answer generation



S: historical Semantic-selected; * omitted. Orange labels: omitted Question top5 in L0-15. P-1 excluded. L31 is observational only.

Omitted whole-group Top5 by scope (all32 heads averaged, no single-head maximum ranking):

L8-11

Question: a @600 0.164; the @584 0.157; the @587 0.139; Is @583 0.135; is @597 0.133

Instruction: word @610 0.078; Answer @604 0.066; . @613 0.061; the @605 0.059; question @606 0.055

Assistant/template: [space] @581 0.318; [NL] @582 0.216; ASSISTANT @614 0.112; [NL] @603 0.085

L0-15

Question: the @584 0.252; Is @583 0.230; the @587 0.229; a @600 0.210; the @594 0.203

Instruction: the @605 0.143; Answer @604 0.138; question @606 0.135; word @610 0.134; . @613 0.131

Assistant/template: [space] @581 0.330; [NL] @582 0.267; ASSISTANT @614 0.166; [NL] @603 0.155

L16-19

Question: the @584 0.241; a @600 0.235; the @594 0.204; is @597 0.195; Is @583 0.175

Instruction: Answer @604 0.068; the @605 0.057; question @606 0.051; or @611 0.048; using @607 0.048

Assistant/template: [space] @581 0.200; ASSISTANT @614 0.121; [NL] @582 0.113; [NL] @603 0.061

L31 R outputs cannot propagate onward to the first-answer P-1 position in the answer-image-blocked causal graph. High L31 mass is not evidence of functional necessity. All answers shown are historical, not Native Full generation in this run.

Is the car to the left or to the right of the train car?



Gold: right
Historical All-Writer: Right
Historical Semantic-k: Left

P-1 is separate (not in R ranking):
L8-11: M=0.0409
L0-15: M=0.1209
L16-19: M=0.0964
L0-31: M=0.0944

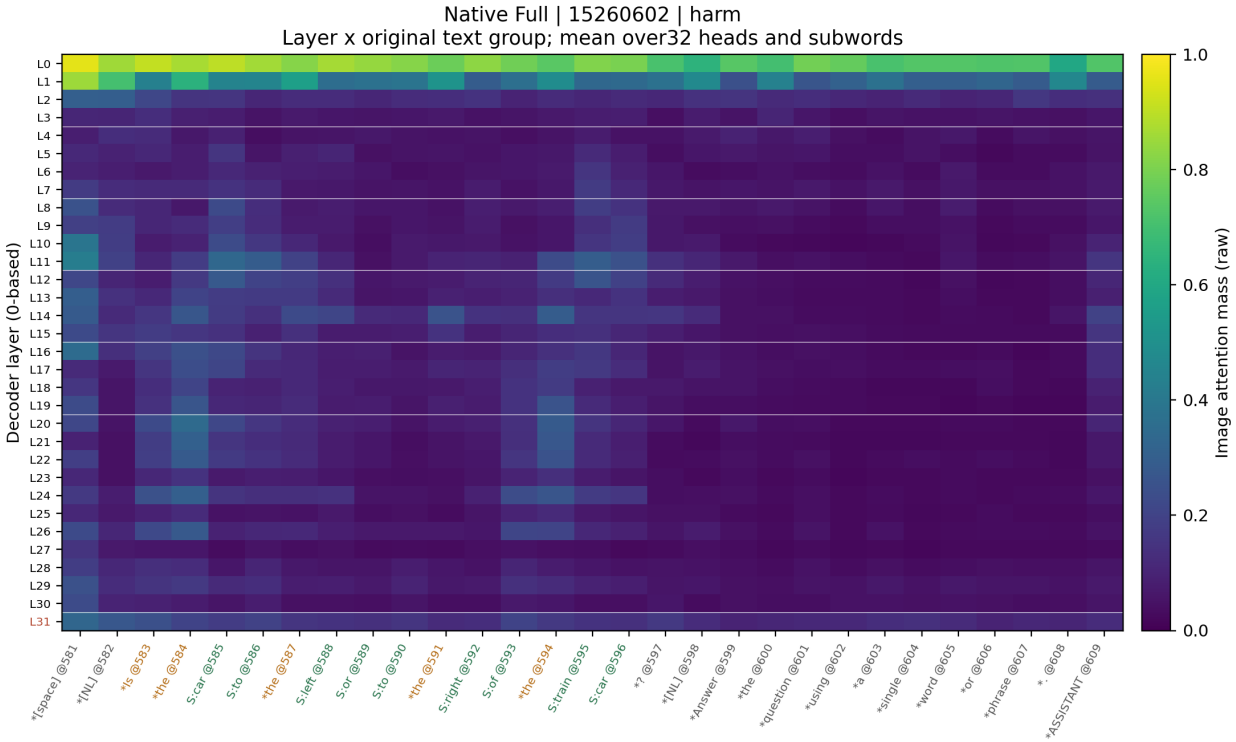
Question groups below: L0-15 averages over all heads/layers/subwords. M=Mass, E=Excess, En=mean per-head Enrichment. Green=historical selected; gray=omitted; orange border=relative omitted top5, not a causal threshold.

Is @583 n=1 M 0.1902 E -0.7961 En 0.0259	the @584 n=1 M 0.2090 E -0.7756 En 0.0250	car @585 n=1 M 0.2395 E -0.7434 En 0.0334	to @586 n=1 M 0.1931 E -0.7881 En 0.0271	the @587 n=1 M 0.1910 E -0.7886 En 0.0257	left @588 n=1 M 0.1611 E -0.8168 En 0.0260
or @589 n=1 M 0.1310 E -0.8453 En 0.0230	to @590 n=1 M 0.1369 E -0.8377 En 0.0235	the @591 n=1 M 0.1576 E -0.8154 En 0.0291	right @592 n=1 M 0.1432 E -0.8281 En 0.0249	of @593 n=1 M 0.1381 E -0.8316 En 0.0242	the @594 n=1 M 0.1772 E -0.7909 En 0.0273
train @595 n=1 M 0.2054 E -0.7610 En 0.0372	car @596 n=1 M 0.1898 E -0.7751 En 0.0302	? @597 n=1 M 0.1393 E -0.8240 En 0.0237			

Other R groups (Instruction and Assistant/template); all historically omitted. Exact subwords and all-layer metrics are preserved in CSV.

Group / positions	Scope	Mass	Excess	Enrich.
[space] @581	Assistant/template	0.3106	-0.6791	0.0570
[NL] @582	Assistant/template	0.2247	-0.7633	0.0253
[NL] @598	Assistant/template	0.1369	-0.8247	0.0246
Answer @599	Instruction	0.1144	-0.8456	0.0194
the @600	Instruction	0.1208	-0.8376	0.0220
question @601	Instruction	0.1117	-0.8451	0.0244
using @602	Instruction	0.1022	-0.8531	0.0231
a @603	Instruction	0.1064	-0.8473	0.0224
single @604	Instruction	0.1007	-0.8514	0.0229
word @605	Instruction	0.1127	-0.8378	0.0238
or @606	Instruction	0.0977	-0.8512	0.0226
phrase @607	Instruction	0.1026	-0.8448	0.0229
. @608	Instruction	0.1092	-0.8366	0.0224
ASSISTANT @609,610,611,612	Assistant/template	0.1499	-0.7920	0.0395

Template [space] immediately after Image is distinct from the ASSISTANT word. Negative Excess and Enrichment below1 mean the observed mass is below the uniform per-key baseline.



S: historical Semantic-selected; * omitted. Orange labels: omitted Question top5 in L0-15. P-1 excluded. L31 is observational only.

Omitted whole-group Top5 by scope (all32 heads averaged, no single-head maximum ranking):

L8-11

Question: the @584 0.115; the @587 0.115; the @594 0.108; Is @583 0.100; ? @597 0.084

Instruction: word @605 0.061; Answer @599 0.051; . @608 0.044; the @600 0.044; question @601 0.039

Assistant/template: [space] @581 0.313; [NL] @582 0.168; ASSISTANT @609 0.097; [NL] @598 0.072

L0-15

Question: the @584 0.209; the @587 0.191; Is @583 0.190; the @594 0.177; the @591 0.158

Instruction: the @600 0.121; Answer @599 0.114; word @605 0.113; question @601 0.112; . @608 0.109

Assistant/template: [space] @581 0.311; [NL] @582 0.225; ASSISTANT @609 0.150; [NL] @598 0.137

L16-19

Question: the @584 0.237; the @594 0.187; Is @583 0.155; the @587 0.117; the @591 0.085

Instruction: Answer @599 0.048; the @600 0.037; question @601 0.037; or @606 0.034; using @602 0.033

Assistant/template: [space] @581 0.213; ASSISTANT @609 0.109; [NL] @582 0.081; [NL] @598 0.058

L31 R outputs cannot propagate onward to the first-answer P-1 position in the answer-image-blocked causal graph. High L31 mass is not evidence of functional necessity. All answers shown are historical, not Native Full generation in this run.

What is common to the shield and the bat?



Gold: color
Historical All-Writer: Color
Historical Semantic-k: Material

P-1 is separate (not in R ranking):
L8-11: M=0.0582
L0-15: M=0.1206
L16-19: M=0.0846
L0-31: M=0.0977

Question groups below: L0-15 averages over all heads/layers/subwords. M=Mass, E=Excess, En=mean per-head Enrichment. Green=historical selected; gray=omitted; orange border=relative omitted top5, not a causal threshold.

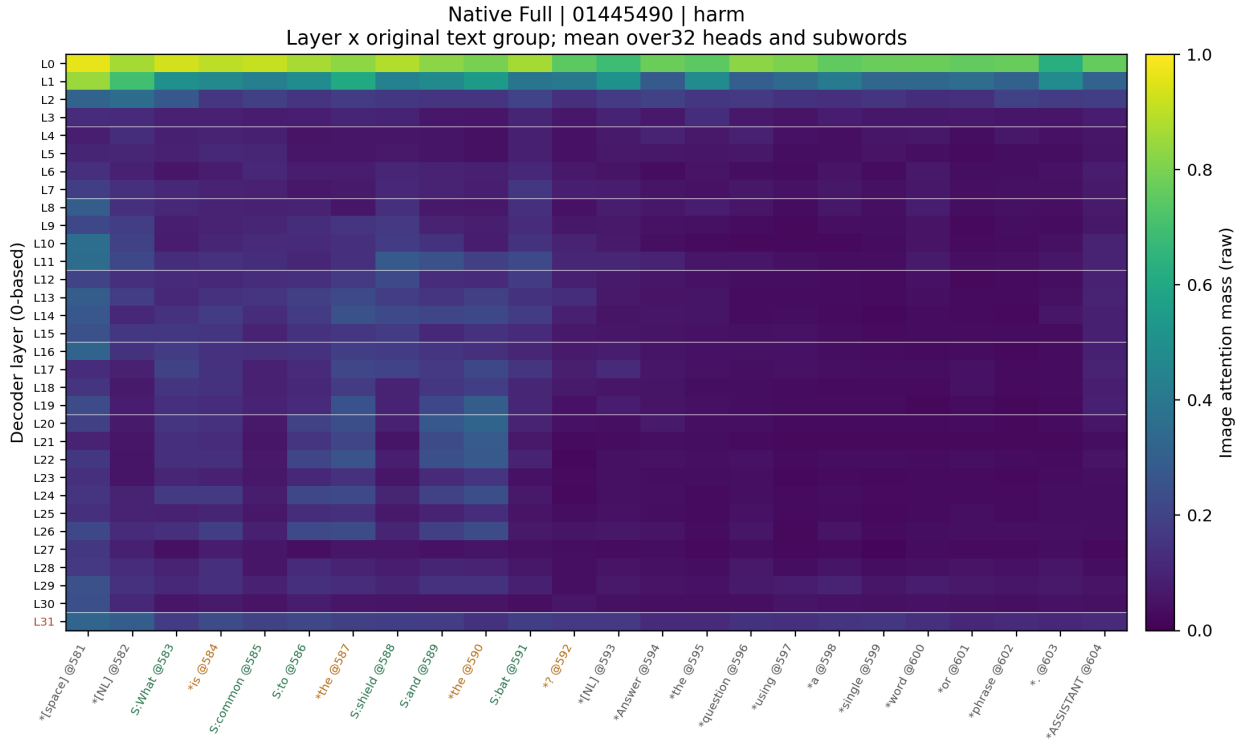
What	is	common	to	the	shield
@583 n=1	@584 n=1	@585 n=1	@586 n=1	@587 n=1	@588 n=1
M 0.1944 E -0.7919	M 0.1906 E -0.7940	M 0.1854 E -0.7975	M 0.1837 E -0.7975	M 0.2042 E -0.7754	M 0.2186 E -0.7594
En 0.0275	En 0.0249	En 0.0278	En 0.0262	En 0.0256	En 0.0312

and	the	bat	?
@589 n=1	@590 n=1	@591 n=1	@592 n=1
M 0.1870 E -0.7893	M 0.1811 E -0.7936	M 0.2037 E -0.7692	M 0.1434 E -0.8280
En 0.0255	En 0.0248	En 0.0304	En 0.0213

Other R groups (Instruction and Assistant/template); all historically omitted. Exact subwords and all-layer metrics are preserved in CSV.

Group / positions	Scope	Mass	Excess	Enrich.
[space] @581	Assistant/template	0.3137	-0.6760	0.0518
[NL] @582	Assistant/template	0.2397	-0.7483	0.0249
[NL] @593	Assistant/template	0.1443	-0.8254	0.0206
Answer @594	Instruction	0.1285	-0.8396	0.0198
the @595	Instruction	0.1364	-0.8301	0.0228
question @596	Instruction	0.1216	-0.8432	0.0255
using @597	Instruction	0.1128	-0.8504	0.0242
a @598	Instruction	0.1174	-0.8442	0.0230
single @599	Instruction	0.1106	-0.8494	0.0234
word @600	Instruction	0.1221	-0.8363	0.0242
or @601	Instruction	0.1076	-0.8493	0.0228
phrase @602	Instruction	0.1119	-0.8433	0.0236
. @603	Instruction	0.1177	-0.8359	0.0227
ASSISTANT @604,605,606,607	Assistant/template	0.1442	-0.8055	0.0295

Template [space] immediately after Image is distinct from the ASSISTANT word. Negative Excess and Enrichment below1 mean the observed mass is below the uniform per-key baseline.



S: historical Semantic-selected; * omitted. Orange labels: omitted Question top5 in L0-15. P-1 excluded. L31 is observational only.

Omitted whole-group Top5 by scope (all32 heads averaged, no single-head maximum ranking):

L8-11

Question: the @587 0.121; is @584 0.111; the @590 0.105; ? @592 0.079

Instruction: word @600 0.064; Answer @594 0.063; the @595 0.055; . @603 0.046; question @596 0.045

Assistant/template: [space] @581 0.307; [NL] @582 0.184; ASSISTANT @604 0.083; [NL] @593 0.082

L0-15

Question: the @587 0.204; is @584 0.191; the @590 0.181; ? @592 0.143

Instruction: the @595 0.136; Answer @594 0.128; word @600 0.122; question @596 0.122; . @603 0.118

Assistant/template: [space] @581 0.314; [NL] @582 0.240; [NL] @593 0.144; ASSISTANT @604 0.144

L16-19

Question: the @590 0.208; the @587 0.206; is @584 0.141; ? @592 0.069

Instruction: Answer @594 0.056; the @595 0.045; or @601 0.042; using @597 0.041; question @596 0.041

Assistant/template: [space] @581 0.207; [NL] @582 0.097; ASSISTANT @604 0.091; [NL] @593 0.083

L31 R outputs cannot propagate onward to the first-answer P-1 position in the answer-image-blocked causal graph. High L31 mass is not evidence of functional necessity. All answers shown are historical, not Native Full generation in this run.

What do the container and the plate have in common?



Gold: color
Historical All-Writer: Color
Historical Semantic-k: Material

P-1 is separate (not in R ranking):
L8-11: M=0.0634
L0-15: M=0.1285
L16-19: M=0.1214
L0-31: M=0.1152

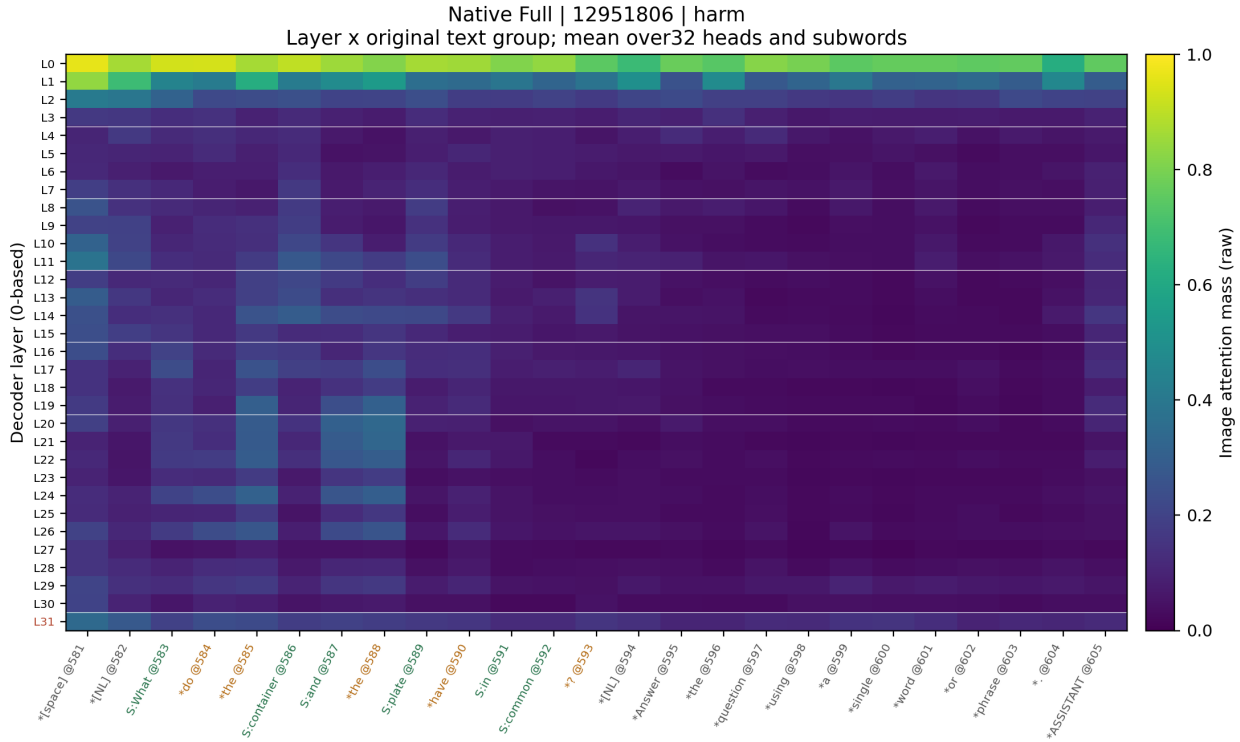
Question groups below: L0-15 averages over all heads/layers/subwords. M=Mass, E=Excess, En=mean per-head Enrichment. Green=historical selected; gray=omitted; orange border=relative omitted top5, not a causal threshold.

What @583 n=1 M 0.2000 E -0.7863 En 0.0275	do @584 n=1 M 0.1925 E -0.7921 En 0.0308	the @585 n=1 M 0.2186 E -0.7643 En 0.0268	container @586 n=1 M 0.2445 E -0.7368 En 0.0354	and @587 n=1 M 0.1924 E -0.7872 En 0.0276	the @588 n=1 M 0.1855 E -0.7924 En 0.0260
plate @589 n=1 M 0.2119 E -0.7644 En 0.0329	have @590 n=1 M 0.1690 E -0.8056 En 0.0269	in @591 n=1 M 0.1505 E -0.8225 En 0.0245	common @592 n=1 M 0.1424 E -0.8290 En 0.0255	? @593 n=1 M 0.1520 E -0.8177 En 0.0230	

Other R groups (Instruction and Assistant/template); all historically omitted. Exact subwords and all-layer metrics are preserved in CSV.

Group / positions	Scope	Mass	Excess	Enrich.
[space] @581	Assistant/template	0.3128	-0.6769	0.0560
[NL] @582	Assistant/template	0.2462	-0.7418	0.0264
[NL] @594	Assistant/template	0.1485	-0.8195	0.0228
Answer @595	Instruction	0.1291	-0.8373	0.0208
the @596	Instruction	0.1373	-0.8275	0.0238
question @597	Instruction	0.1229	-0.8403	0.0261
using @598	Instruction	0.1134	-0.8482	0.0247
a @599	Instruction	0.1193	-0.8407	0.0235
single @600	Instruction	0.1114	-0.8471	0.0238
word @601	Instruction	0.1232	-0.8336	0.0247
or @602	Instruction	0.1077	-0.8476	0.0235
phrase @603	Instruction	0.1130	-0.8407	0.0244
. @604	Instruction	0.1226	-0.8294	0.0237
ASSISTANT @605,606,607,608	Assistant/template	0.1619	-0.7863	0.0340

Template [space] immediately after Image is distinct from the ASSISTANT word. Negative Excess and Enrichment below1 mean the observed mass is below the uniform per-key baseline.



S: historical Semantic-selected; * omitted. Orange labels: omitted Question top5 in L0-15. P-1 excluded. L31 is observational only.

Omitted whole-group Top5 by scope (all32 heads averaged, no single-head maximum ranking):

L8-11

Question: the @585 0.135; do @584 0.119; have @590 0.099; the @588 0.097; ? @593 0.090

Instruction: word @601 0.065; Answer @595 0.063; the @596 0.055; . @604 0.052; a @599 0.046

Assistant/template: [space] @581 0.288; [NL] @582 0.186; ASSISTANT @605 0.116; [NL] @594 0.085

L0-15

Question: the @585 0.219; do @584 0.192; the @588 0.186; have @590 0.169; ? @593 0.152

Instruction: the @596 0.137; Answer @595 0.129; word @601 0.123; question @597 0.123; . @604 0.123

Assistant/template: [space] @581 0.313; [NL] @582 0.246; ASSISTANT @605 0.162; [NL] @594 0.149

L16-19

Question: the @585 0.231; the @588 0.220; have @590 0.122; do @584 0.104; ? @593 0.068

Instruction: Answer @595 0.049; the @596 0.040; or @602 0.038; question @597 0.035; . @604 0.034

Assistant/template: [space] @581 0.175; ASSISTANT @605 0.111; [NL] @582 0.093; [NL] @594 0.074

L31 R outputs cannot propagate onward to the first-answer P-1 position in the answer-image-blocked causal graph. High L31 mass is not evidence of functional necessity. All answers shown are historical, not Native Full generation in this run.

Do you see a door or a window that are open?



Gold: yes
Historical All-Writer: Yes
Historical Semantic-k: Door

P-1 is separate (not in R ranking):
L8-11: M=0.0452
L0-15: M=0.1232
L16-19: M=0.0523
L0-31: M=0.0895

Question groups below: L0-15 averages over all heads/layers/subwords. M=Mass, E=Excess, En=mean per-head Enrichment. Green=historical selected; gray=omitted; orange border=relative omitted top5, not a causal threshold.

Do @583 n=1 M 0.1983 E -0.7880 En 0.0263	you @584 n=1 M 0.1725 E -0.8121 En 0.0267	see @585 n=1 M 0.1668 E -0.8161 En 0.0286	a @586 n=1 M 0.1924 E -0.7888 En 0.0266	door @587 n=1 M 0.2259 E -0.7537 En 0.0400	or @588 n=1 M 0.1398 E -0.8381 En 0.0255
a @589 n=1 M 0.1512 E -0.8251 En 0.0251	window @590 n=1 M 0.1848 E -0.7898 En 0.0320	that @591 n=1 M 0.1438 E -0.8292 En 0.0261	are @592 n=1 M 0.1521 E -0.8192 En 0.0269	open @593 n=1 M 0.1714 E -0.7983 En 0.0344	? @594 n=1 M 0.1304 E -0.8376 En 0.0233

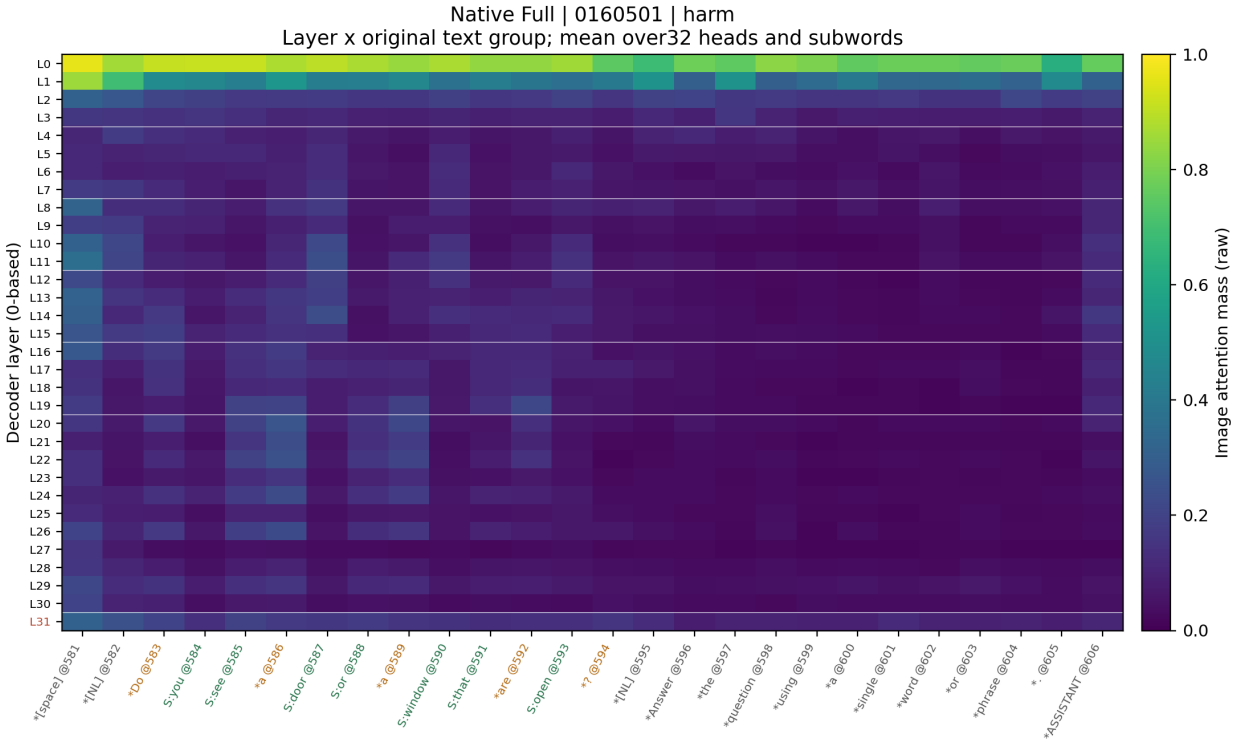
Other R groups (Instruction and Assistant/template); all historically omitted. Exact subwords and all-layer metrics are preserved in CSV.

Group / positions	Scope	Mass	Excess	Enrich.
[space] @581	Assistant/template	0.3175	-0.6722	0.0509
[NL] @582	Assistant/template	0.2355	-0.7524	0.0247
[NL] @595	Assistant/template	0.1399	-0.8265	0.0232
Answer @596	Instruction	0.1244	-0.8404	0.0221
the @597	Instruction	0.1335	-0.8297	0.0252
question @598	Instruction	0.1206	-0.8410	0.0278
using @599	Instruction	0.1114	-0.8486	0.0264
a @600	Instruction	0.1148	-0.8436	0.0244
single @601	Instruction	0.1095	-0.8473	0.0254
word @602	Instruction	0.1193	-0.8359	0.0263
or @603	Instruction	0.1060	-0.8476	0.0246
phrase @604	Instruction	0.1114	-0.8407	0.0260
. @605	Instruction	0.1163	-0.8342	0.0243
ASSISTANT @606,607,608,609	Assistant/template	0.1662	-0.7804	0.0426

Template [space] immediately after Image is distinct from the ASSISTANT word. Negative Excess and Enrichment below1 mean the observed mass is below the uniform per-key baseline.

09 | 0160501 | LogicalObj | harm

NATIVE FULL OBSERVATION | layer heatmap and omitted Top5 | no new answer generation



S: historical Semantic-selected; * omitted. Orange labels: omitted Question top5 in L0-15. P-1 excluded. L31 is observational only.

Omitted whole-group Top5 by scope (all32 heads averaged, no single-head maximum ranking):

L8-11

Question: a @586 0.114; Do @583 0.105; a @589 0.080; are @592 0.065; ? @594 0.056

Instruction: word @602 0.056; Answer @596 0.045; the @597 0.044; . @605 0.042; question @598 0.038

Assistant/template: [space] @581 0.294; [NL] @582 0.181; ASSISTANT @606 0.118; [NL] @595 0.064

L0-15

Question: Do @583 0.198; a @586 0.192; are @592 0.152; a @589 0.151; ? @594 0.130

Instruction: the @597 0.133; Answer @596 0.124; question @598 0.121; word @602 0.119; . @605 0.116

Assistant/template: [space] @581 0.317; [NL] @582 0.236; ASSISTANT @606 0.166; [NL] @595 0.140

L16-19

Question: a @586 0.165; are @592 0.147; Do @583 0.135; a @589 0.124; ? @594 0.064

Instruction: Answer @596 0.044; or @603 0.033; the @597 0.032; question @598 0.031; using @599 0.028

Assistant/template: [space] @581 0.183; ASSISTANT @606 0.102; [NL] @582 0.084; [NL] @595 0.051

L31 R outputs cannot propagate onward to the first-answer P-1 position in the answer-image-blocked causal graph. High L31 mass is not evidence of functional necessity. All answers shown are historical, not Native Full generation in this run.

Are there any TVs or laptops that are on?



Gold: yes
Historical All-Writer: Yes
Historical Semantic-k: No

P-1 is separate (not in R ranking):
L8-11: M=0.0655
L0-15: M=0.1372
L16-19: M=0.0665
L0-31: M=0.1016

Question groups below: L0-15 averages over all heads/layers/subwords. M=Mass, E=Excess, En=mean per-head Enrichment. Green=historical selected; gray=omitted; orange border=relative omitted top5, not a causal threshold.

Are	there	any	TVs	or	laptops
@583 n=1 M 0.2027 E -0.7836 En 0.0265	@584 n=1 M 0.1944 E -0.7903 En 0.0292	@585 n=1 M 0.1910 E -0.7920 En 0.0286	@586 n=2 M 0.2154 E -0.7650 En 0.0331	@588 n=1 M 0.1392 E -0.8387 En 0.0257	@589 n=3 M 0.1978 E -0.7768 En 0.0330

that	are	on	?
@592 n=1 M 0.1504 E -0.8209 En 0.0267	@593 n=1 M 0.1664 E -0.8033 En 0.0296	@594 n=1 M 0.1758 E -0.7922 En 0.0318	@595 n=1 M 0.1704 E -0.7960 En 0.0268

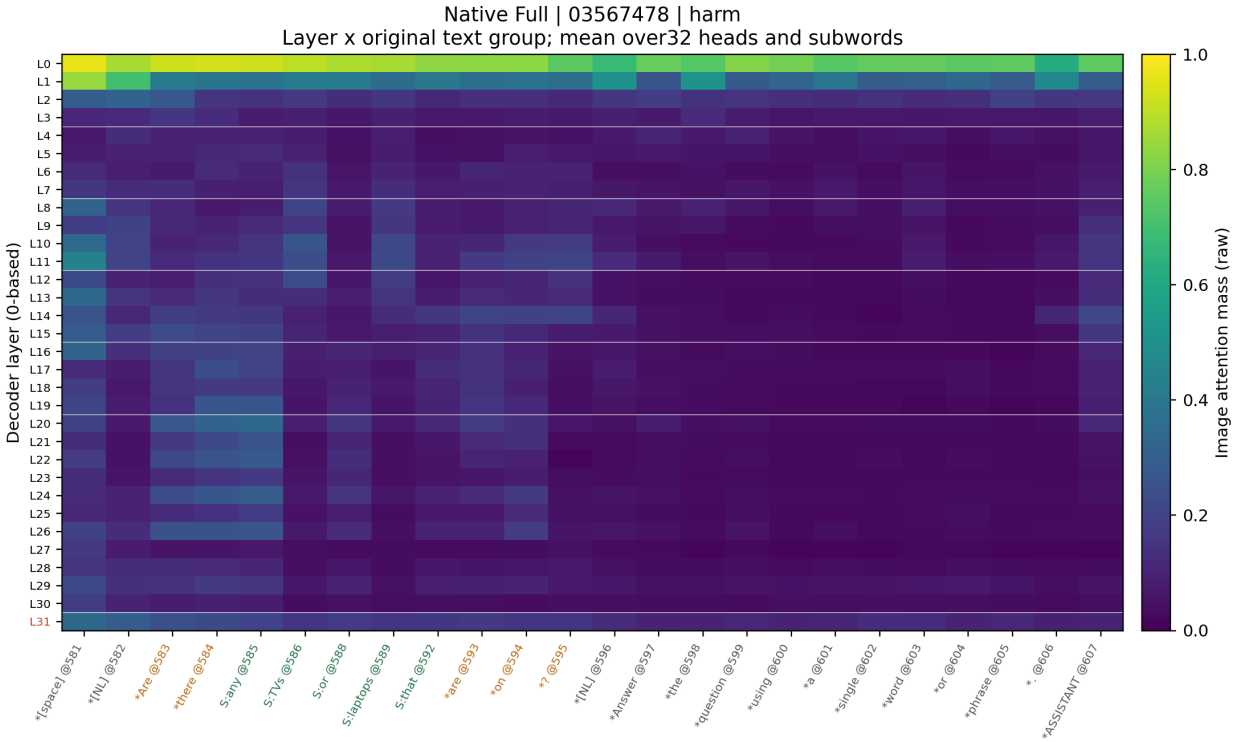
Other R groups (Instruction and Assistant/template); all historically omitted. Exact subwords and all-layer metrics are preserved in CSV.

Group / positions	Scope	Mass	Excess	Enrich.
[space] @581	Assistant/template	0.3155	-0.6742	0.0612
[NL] @582	Assistant/template	0.2328	-0.7552	0.0267
[NL] @596	Assistant/template	0.1452	-0.8196	0.0253
Answer @597	Instruction	0.1209	-0.8423	0.0213
the @598	Instruction	0.1310	-0.8306	0.0245
question @599	Instruction	0.1167	-0.8433	0.0264
using @600	Instruction	0.1082	-0.8502	0.0248
a @601	Instruction	0.1123	-0.8445	0.0233
single @602	Instruction	0.1043	-0.8509	0.0239
word @603	Instruction	0.1188	-0.8348	0.0252
or @604	Instruction	0.1022	-0.8499	0.0239
phrase @605	Instruction	0.1068	-0.8436	0.0248
. @606	Instruction	0.1199	-0.8290	0.0242
ASSISTANT @607,608,609,610	Assistant/template	0.1709	-0.7742	0.0509

Template [space] immediately after Image is distinct from the ASSISTANT word. Negative Excess and Enrichment below1 mean the observed mass is below the uniform per-key baseline.

10 | 03567478 | LogicalObj | harm

NATIVE FULL OBSERVATION | layer heatmap and omitted Top5 | no new answer generation



S: historical Semantic-selected; * omitted. Orange labels: omitted Question top5 in L0-15. P-1 excluded. L31 is observational only.

Omitted whole-group Top5 by scope (all32 heads averaged, no single-head maximum ranking):

L8-11

Question: ? @595 0.147; on @594 0.136; Are @583 0.112; there @584 0.106; are @593 0.105

Instruction: word @603 0.071; Answer @597 0.057; . @606 0.055; the @598 0.055; question @599 0.044

Assistant/template: [space] @581 0.321; [NL] @582 0.188; ASSISTANT @607 0.135; [NL] @596 0.095

L0-15

Question: Are @583 0.203; there @584 0.194; on @594 0.176; ? @595 0.170; are @593 0.166

Instruction: the @598 0.131; Answer @597 0.121; . @606 0.120; word @603 0.119; question @599 0.117

Assistant/template: [space] @581 0.315; [NL] @582 0.233; ASSISTANT @607 0.171; [NL] @596 0.145

L16-19

Question: there @584 0.217; Are @583 0.163; are @593 0.145; on @594 0.099; ? @595 0.045

Instruction: Answer @597 0.043; question @599 0.034; the @598 0.033; or @604 0.033; using @600 0.029

Assistant/template: [space] @581 0.208; ASSISTANT @607 0.097; [NL] @582 0.090; [NL] @596 0.064

L31 R outputs cannot propagate onward to the first-answer P-1 position in the answer-image-blocked causal graph. High L31 mass is not evidence of functional necessity. All answers shown are historical, not Native Full generation in this run.

On which side is the small container?



Gold: right
Historical All-Writer: Right
Historical Semantic-k: Left

P-1 is separate (not in R ranking):
L8-11: M=0.0824
L0-15: M=0.1600
L16-19: M=0.0997
L0-31: M=0.1212

Question groups below: L0-15 averages over all heads/layers/subwords. M=Mass, E=Excess, En=mean per-head Enrichment. Green=historical selected; gray=omitted; orange border=relative omitted top5, not a causal threshold.

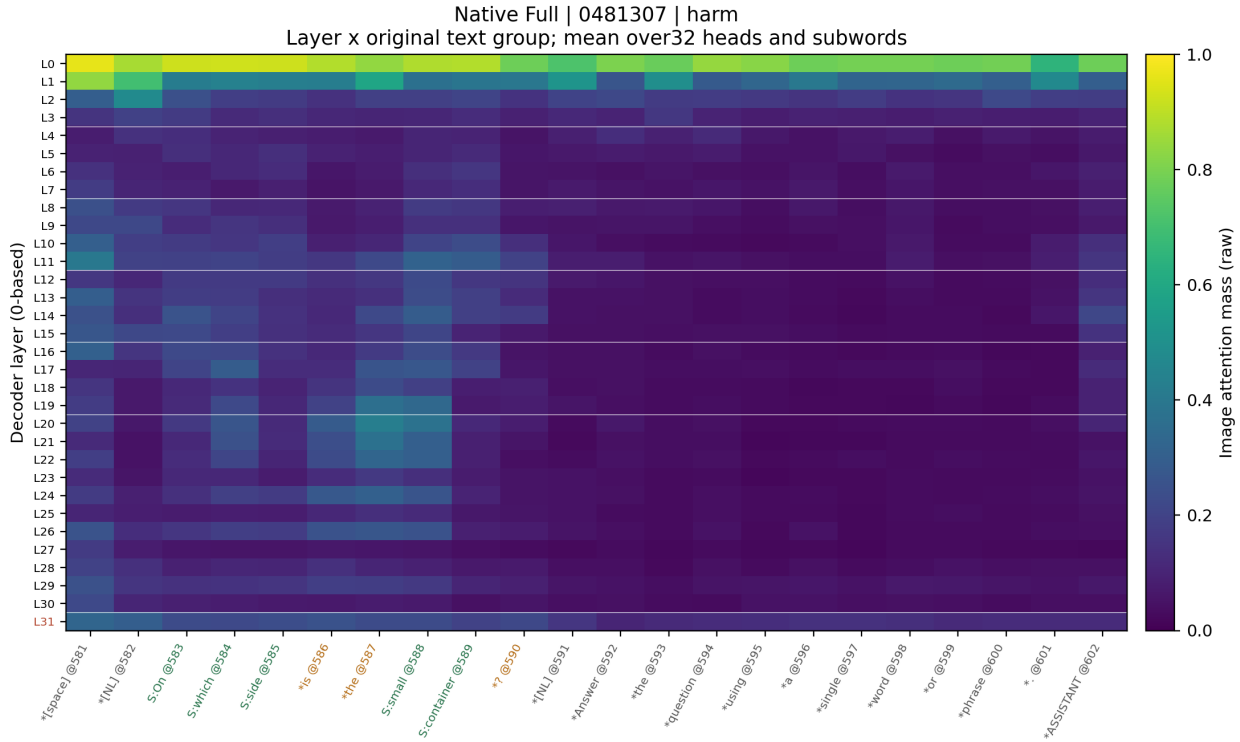
On	which	side	is	the	small
@583 n=1 M 0.2307 E -0.7556 En 0.0279	@584 n=1 M 0.2145 E -0.7701 En 0.0295	@585 n=1 M 0.2061 E -0.7768 En 0.0318	@586 n=1 M 0.1713 E -0.8099 En 0.0263	@587 n=1 M 0.1999 E -0.7797 En 0.0271	@588 n=1 M 0.2369 E -0.7411 En 0.0336

container	?
@589 n=1 M 0.2238 E -0.7525 En 0.0341	@590 n=1 M 0.1648 E -0.8099 En 0.0275

Other R groups (Instruction and Assistant/template); all historically omitted. Exact subwords and all-layer metrics are preserved in CSV.

Group / positions	Scope	Mass	Excess	Enrich.
[space] @581	Assistant/template	0.3053	-0.6844	0.0538
[NL] @582	Assistant/template	0.2535	-0.7345	0.0268
[NL] @591	Assistant/template	0.1462	-0.8267	0.0226
Answer @592	Instruction	0.1340	-0.8373	0.0209
the @593	Instruction	0.1415	-0.8282	0.0242
question @594	Instruction	0.1273	-0.8408	0.0273
using @595	Instruction	0.1187	-0.8478	0.0262
a @596	Instruction	0.1222	-0.8426	0.0246
single @597	Instruction	0.1156	-0.8477	0.0254
word @598	Instruction	0.1268	-0.8348	0.0261
or @599	Instruction	0.1121	-0.8479	0.0242
phrase @600	Instruction	0.1164	-0.8420	0.0251
. @601	Instruction	0.1256	-0.8312	0.0235
ASSISTANT @602,603,604,605	Assistant/template	0.1720	-0.7809	0.0570

Template [space] immediately after Image is distinct from the ASSISTANT word. Negative Excess and Enrichment below1 mean the observed mass is below the uniform per-key baseline.



S: historical Semantic-selected; * omitted. Orange labels: omitted Question top5 in L0-15. P-1 excluded. L31 is observational only.

Omitted whole-group Top5 by scope (all32 heads averaged, no single-head maximum ranking):

L8-11

Question: the @587 0.126; ? @590 0.120; is @586 0.098

Instruction: word @598 0.068; Answer @592 0.062; . @601 0.061; the @593 0.054; question @594 0.047

Assistant/template: [space] @581 0.291; [NL] @582 0.193; ASSISTANT @602 0.111; [NL] @591 0.071

L0-15

Question: the @587 0.200; is @586 0.171; ? @590 0.165

Instruction: the @593 0.142; Answer @592 0.134; question @594 0.127; word @598 0.127; . @601 0.126

Assistant/template: [space] @581 0.305; [NL] @582 0.253; ASSISTANT @602 0.172; [NL] @591 0.146

L16-19

Question: the @587 0.256; is @586 0.148; ? @590 0.071

Instruction: Answer @592 0.043; question @594 0.037; or @599 0.037; the @593 0.035; using @595 0.033

Assistant/template: [space] @581 0.187; ASSISTANT @602 0.101; [NL] @582 0.098; [NL] @591 0.047

L31 R outputs cannot propagate onward to the first-answer P-1 position in the answer-image-blocked causal graph. High L31 mass is not evidence of functional necessity. All answers shown are historical, not Native Full generation in this run.

On which side is the can?



Gold: right
Historical All-Writer: Right
Historical Semantic-k: Left

P-1 is separate (not in R ranking):
L8-11: M=0.0829
L0-15: M=0.1536
L16-19: M=0.0896
L0-31: M=0.1165

Question groups below: L0-15 averages over all heads/layers/subwords. M=Mass, E=Excess, En=mean per-head Enrichment. Green=historical selected; gray=omitted; orange border=relative omitted top5, not a causal threshold.

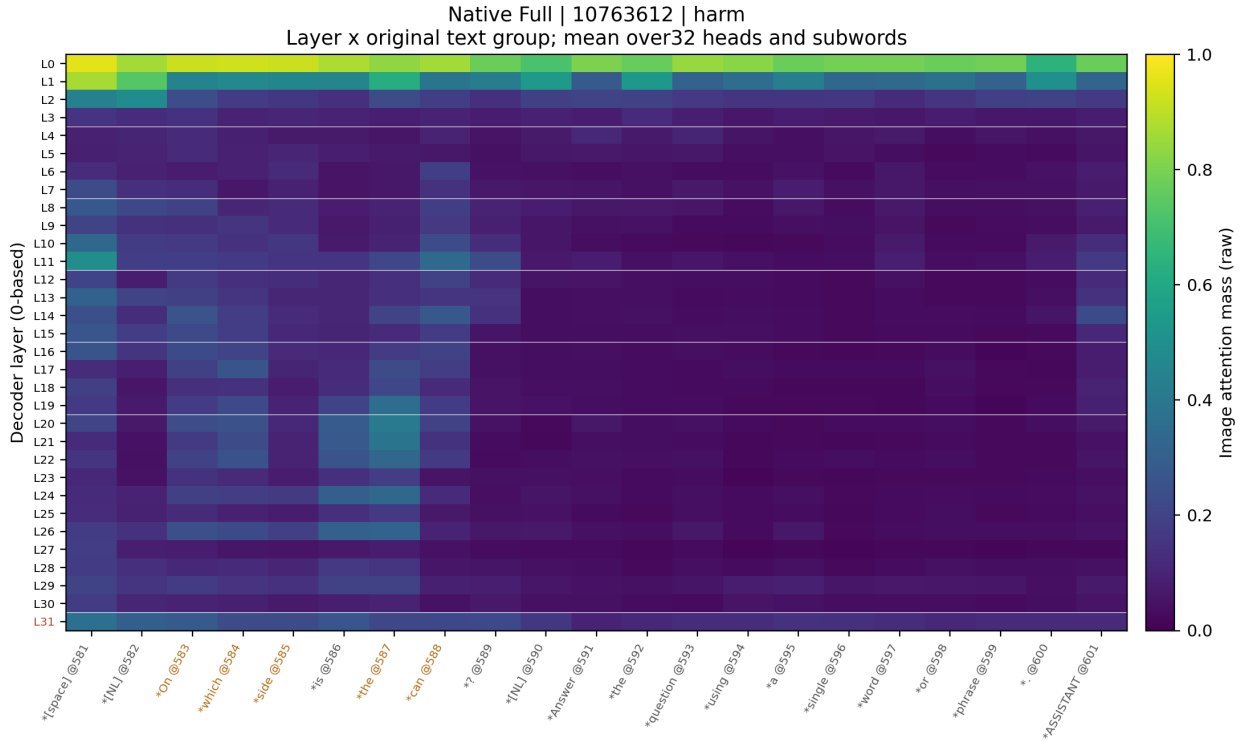
On	which	side	is	the	can
@583 n=1	@584 n=1	@585 n=1	@586 n=1	@587 n=1	@588 n=1
M 0.2319 E -0.7544	M 0.2017 E -0.7829	M 0.1912 E -0.7918	M 0.1642 E -0.8170	M 0.1952 E -0.7844	M 0.2339 E -0.7440
En 0.0272	En 0.0284	En 0.0305	En 0.0253	En 0.0265	En 0.0359

?
@589 n=1
M 0.1650 E -0.8113
En 0.0249

Other R groups (Instruction and Assistant/template); all historically omitted. Exact subwords and all-layer metrics are preserved in CSV.

Group / positions	Scope	Mass	Excess	Enrich.
[space] @581	Assistant/template	0.3302	-0.6595	0.0590
[NL] @582	Assistant/template	0.2473	-0.7407	0.0267
[NL] @590	Assistant/template	0.1385	-0.8362	0.0223
Answer @591	Instruction	0.1274	-0.8456	0.0197
the @592	Instruction	0.1373	-0.8340	0.0231
question @593	Instruction	0.1243	-0.8454	0.0258
using @594	Instruction	0.1155	-0.8525	0.0247
a @595	Instruction	0.1213	-0.8452	0.0233
single @596	Instruction	0.1117	-0.8532	0.0239
word @597	Instruction	0.1235	-0.8397	0.0247
or @598	Instruction	0.1095	-0.8522	0.0232
phrase @599	Instruction	0.1113	-0.8487	0.0236
. @600	Instruction	0.1234	-0.8350	0.0226
ASSISTANT @601,602,603,604	Assistant/template	0.1681	-0.7864	0.1432

Template [space] immediately after Image is distinct from the ASSISTANT word. Negative Excess and Enrichment below1 mean the observed mass is below the uniform per-key baseline.



S: historical Semantic-selected; * omitted. Orange labels: omitted Question top5 in L0-15. P-1 excluded. L31 is observational only.

Omitted whole-group Top5 by scope (all32 heads averaged, no single-head maximum ranking):

L8-11

Question: can @588 0.231; On @583 0.171; which @584 0.143; side @585 0.139; ? @589 0.136

Instruction: word @597 0.069; . @600 0.059; Answer @591 0.057; the @592 0.049; question @593 0.044

Assistant/template: [space] @581 0.329; [NL] @582 0.181; ASSISTANT @601 0.115; [NL] @590 0.067

L0-15

Question: can @588 0.234; On @583 0.232; which @584 0.202; the @587 0.195; side @585 0.191

Instruction: the @592 0.137; Answer @591 0.127; question @593 0.124; word @597 0.123; . @600 0.123

Assistant/template: [space] @581 0.330; [NL] @582 0.247; ASSISTANT @601 0.168; [NL] @590 0.138

L16-19

Question: the @587 0.245; which @584 0.207; On @583 0.181; can @588 0.169; is @586 0.141

Instruction: Answer @591 0.038; or @598 0.036; question @593 0.034; the @592 0.033; using @594 0.031

Assistant/template: [space] @581 0.186; [NL] @582 0.093; ASSISTANT @601 0.084; [NL] @590 0.041

L31 R outputs cannot propagate onward to the first-answer P-1 position in the answer-image-blocked causal graph. High L31 mass is not evidence of functional necessity. All answers shown are historical, not Native Full generation in this run.

Is the mud dry or wet?



Gold: wet
Historical All-Writer: Dry
Historical Semantic-k: Wet

P-1 is separate (not in R ranking):
L8-11: M=0.0944
L0-15: M=0.1510
L16-19: M=0.0973
L0-31: M=0.1119

Question groups below: L0-15 averages over all heads/layers/subwords. M=Mass, E=Excess, En=mean per-head Enrichment. Green=historical selected; gray=omitted; orange border=relative omitted top5, not a causal threshold.

Is	the	mud	dry	or	wet
@583 n=1	@584 n=1	@585 n=1	@586 n=1	@587 n=1	@588 n=1
M 0.2075 E -0.7788	M 0.2297 E -0.7549	M 0.2622 E -0.7207	M 0.2031 E -0.7782	M 0.1541 E -0.8255	M 0.1739 E -0.8041
En 0.0263	En 0.0260	En 0.0335	En 0.0291	En 0.0244	En 0.0286

?
@589 n=1
M 0.1629 E -0.8133
En 0.0222

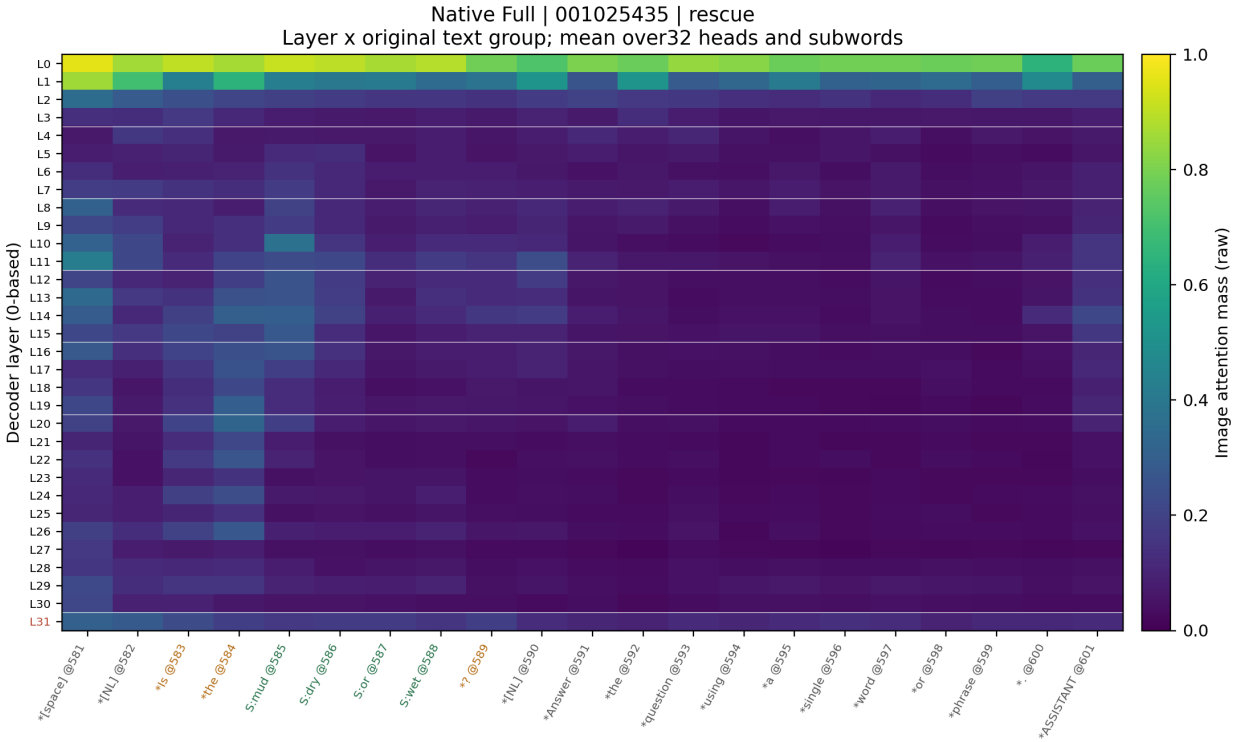
Other R groups (Instruction and Assistant/template); all historically omitted. Exact subwords and all-layer metrics are preserved in CSV.

Group / positions	Scope	Mass	Excess	Enrich.
[space] @581	Assistant/template	0.3178	-0.6719	0.0557
[NL] @582	Assistant/template	0.2363	-0.7517	0.0257
[NL] @590	Assistant/template	0.1843	-0.7903	0.0236
Answer @591	Instruction	0.1368	-0.8362	0.0200
the @592	Instruction	0.1465	-0.8248	0.0235
question @593	Instruction	0.1277	-0.8420	0.0265
using @594	Instruction	0.1180	-0.8501	0.0254
a @595	Instruction	0.1254	-0.8410	0.0238
single @596	Instruction	0.1146	-0.8502	0.0242
word @597	Instruction	0.1304	-0.8328	0.0250
or @598	Instruction	0.1102	-0.8514	0.0232
phrase @599	Instruction	0.1163	-0.8437	0.0239
. @600	Instruction	0.1324	-0.8260	0.0228
ASSISTANT @601,602,603,604	Assistant/template	0.1760	-0.7784	0.0349

Template [space] immediately after Image is distinct from the ASSISTANT word. Negative Excess and Enrichment below1 mean the observed mass is below the uniform per-key baseline.

13 | 001025435 | ChooseAttr | rescue

NATIVE FULL OBSERVATION | layer heatmap and omitted Top5 | no new answer generation



S: historical Semantic-selected; * omitted. Orange labels: omitted Question top5 in L0-15. P-1 excluded. L31 is observational only.

Omitted whole-group Top5 by scope (all32 heads averaged, no single-head maximum ranking):

L8-11

Question: the @584 0.137; Is @583 0.113; ? @589 0.110

Instruction: word @597 0.081; Answer @591 0.073; the @592 0.068; . @600 0.067; question @593 0.053

Assistant/template: [space] @581 0.315; [NL] @582 0.183; [NL] @590 0.142; ASSISTANT @601 0.128

L0-15

Question: the @584 0.230; Is @583 0.208; ? @589 0.163

Instruction: the @592 0.147; Answer @591 0.137; . @600 0.132; word @597 0.130; question @593 0.128

Assistant/template: [space] @581 0.318; [NL] @582 0.236; [NL] @590 0.184; ASSISTANT @601 0.176

L16-19

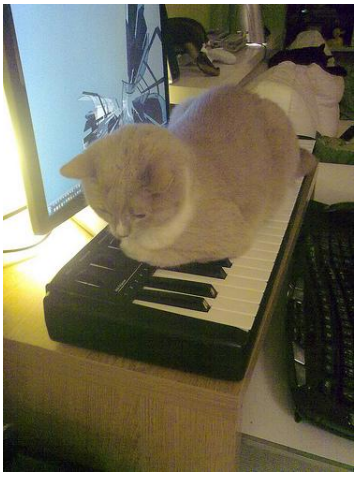
Question: the @584 0.252; Is @583 0.159; ? @589 0.072

Instruction: Answer @591 0.060; the @592 0.041; or @598 0.039; question @593 0.038; . @600 0.037

Assistant/template: [space] @581 0.196; ASSISTANT @601 0.105; [NL] @582 0.091; [NL] @590 0.077

L31 R outputs cannot propagate onward to the first-answer P-1 position in the answer-image-blocked causal graph. High L31 mass is not evidence of functional necessity. All answers shown are historical, not Native Full generation in this run.

What type of furniture is not white, the drawer or the chair?



Gold: drawer
Historical All-Writer: Chair
Historical Semantic-k: Drawer

P-1 is separate (not in R ranking):
L8-11: M=0.0450
L0-15: M=0.1238
L16-19: M=0.0779
L0-31: M=0.0994

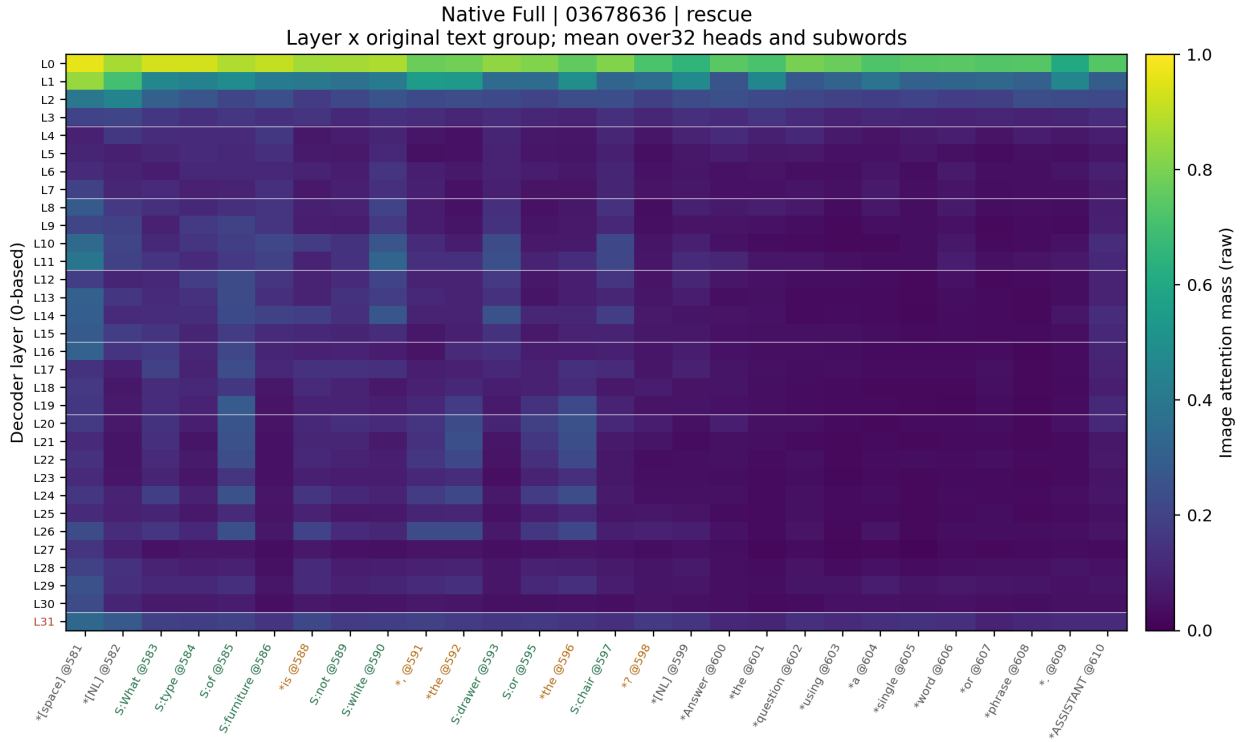
Question groups below: L0-15 averages over all heads/layers/subwords. M=Mass, E=Excess, En=mean per-head Enrichment. Green=historical selected; gray=omitted; orange border=relative omitted top5, not a causal threshold.

What @583 n=1 M 0.2061 E -0.7802 En 0.0274	type @584 n=1 M 0.2085 E -0.7762 En 0.0305	of @585 n=1 M 0.2330 E -0.7500 En 0.0285	furniture @586 n=2 M 0.2228 E -0.7577 En 0.0320	is @588 n=1 M 0.1766 E -0.8013 En 0.0251	not @589 n=1 M 0.1784 E -0.7978 En 0.0267
white @590 n=1 M 0.2450 E -0.7296 En 0.0449	, @591 n=1 M 0.1706 E -0.8024 En 0.0256	the @592 n=1 M 0.1617 E -0.8097 En 0.0248	drawer @593 n=2 M 0.2114 E -0.7575 En 0.0323	or @595 n=1 M 0.1430 E -0.8234 En 0.0252	the @596 n=1 M 0.1574 E -0.8074 En 0.0259
chair @597 n=1 M 0.1963 E -0.7669 En 0.0324	? @598 n=1 M 0.1260 E -0.8356 En 0.0222				

Other R groups (Instruction and Assistant/template); all historically omitted. Exact subwords and all-layer metrics are preserved in CSV.

Group / positions	Scope	Mass	Excess	Enrich.
[space] @581	Assistant/template	0.3262	-0.6635	0.0588
[NL] @582	Assistant/template	0.2506	-0.7374	0.0267
[NL] @599	Assistant/template	0.1521	-0.8079	0.0238
Answer @600	Instruction	0.1342	-0.8242	0.0217
the @601	Instruction	0.1381	-0.8187	0.0243
question @602	Instruction	0.1262	-0.8290	0.0267
using @603	Instruction	0.1165	-0.8372	0.0254
a @604	Instruction	0.1207	-0.8314	0.0246
single @605	Instruction	0.1131	-0.8374	0.0252
word @606	Instruction	0.1258	-0.8231	0.0262
or @607	Instruction	0.1110	-0.8364	0.0247
phrase @608	Instruction	0.1162	-0.8296	0.0253
. @609	Instruction	0.1229	-0.8214	0.0245
ASSISTANT @610,611,612,613	Assistant/template	0.1578	-0.7826	0.0341

Template [space] immediately after Image is distinct from the ASSISTANT word. Negative Excess and Enrichment below1 mean the observed mass is below the uniform per-key baseline.



S: historical Semantic-selected; * omitted. Orange labels: omitted Question top5 in L0-15. P-1 excluded. L31 is observational only.

Omitted whole-group Top5 by scope (all32 heads averaged, no single-head maximum ranking):

L8-11

Question: is @588 0.110; , @591 0.103; the @596 0.084; the @592 0.080; ? @598 0.049

Instruction: Answer @600 0.070; word @606 0.066; the @601 0.055; question @602 0.049; . @609 0.047

Assistant/template: [space] @581 0.307; [NL] @582 0.192; ASSISTANT @610 0.102; [NL] @599 0.087

L0-15

Question: is @588 0.177; , @591 0.171; the @592 0.162; the @596 0.157; ? @598 0.126

Instruction: the @601 0.138; Answer @600 0.134; question @602 0.126; word @606 0.126; . @609 0.123

Assistant/template: [space] @581 0.326; [NL] @582 0.251; ASSISTANT @610 0.158; [NL] @599 0.152

L16-19

Question: the @596 0.140; the @592 0.125; is @588 0.113; , @591 0.085; ? @598 0.064

Instruction: Answer @600 0.053; the @601 0.042; or @607 0.037; question @602 0.036; using @603 0.035

Assistant/template: [space] @581 0.197; ASSISTANT @610 0.099; [NL] @582 0.093; [NL] @599 0.070

L31 R outputs cannot propagate onward to the first-answer P-1 position in the answer-image-blocked causal graph. High L31 mass is not evidence of functional necessity. All answers shown are historical, not Native Full generation in this run.

Is the bag to the right or to the left of the man?



Gold: left
Historical All-Writer: Right
Historical Semantic-k: Left

P-1 is separate (not in R ranking):
L8-11: M=0.0432
L0-15: M=0.1358
L16-19: M=0.1009
L0-31: M=0.1046

Question groups below: L0-15 averages over all heads/layers/subwords. M=Mass, E=Excess, En=mean per-head Enrichment. Green=historical selected; gray=omitted; orange border=relative omitted top5, not a causal threshold.

Is @583 n=1 M 0.2163 E -0.7700 En 0.0275	the @584 n=1 M 0.2405 E -0.7441 En 0.0270	bag @585 n=1 M 0.2617 E -0.7213 En 0.0394	to @586 n=1 M 0.2014 E -0.7798 En 0.0280	the @587 n=1 M 0.2115 E -0.7681 En 0.0317	right @588 n=1 M 0.1754 E -0.8025 En 0.0275
or @589 n=1 M 0.1467 E -0.8295 En 0.0245	to @590 n=1 M 0.1540 E -0.8206 En 0.0250	the @591 n=1 M 0.1761 E -0.7969 En 0.0363	left @592 n=1 M 0.1663 E -0.8051 En 0.0267	of @593 n=1 M 0.1603 E -0.8094 En 0.0262	the @594 n=1 M 0.1905 E -0.7775 En 0.0281
man @595 n=1 M 0.2528 E -0.7136 En 0.0431	? @596 n=1 M 0.1475 E -0.8174 En 0.0242				

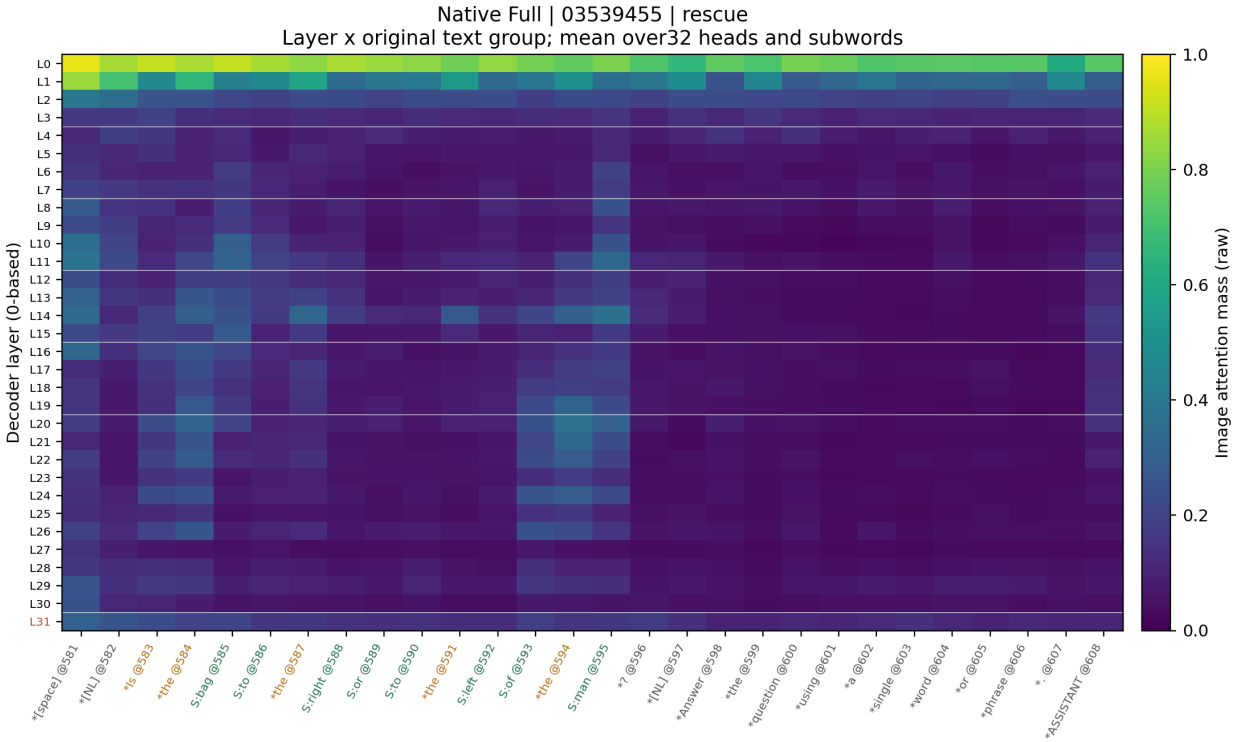
Other R groups (Instruction and Assistant/template); all historically omitted. Exact subwords and all-layer metrics are preserved in CSV.

Group / positions	Scope	Mass	Excess	Enrich.
[space] @581	Assistant/template	0.3339	-0.6557	0.0547
[NL] @582	Assistant/template	0.2502	-0.7378	0.0260
[NL] @597	Assistant/template	0.1489	-0.8143	0.0231
Answer @598	Instruction	0.1311	-0.8305	0.0207
the @599	Instruction	0.1372	-0.8228	0.0236
question @600	Instruction	0.1282	-0.8302	0.0258
using @601	Instruction	0.1189	-0.8379	0.0244
a @602	Instruction	0.1233	-0.8319	0.0237
single @603	Instruction	0.1178	-0.8358	0.0241
word @604	Instruction	0.1288	-0.8233	0.0252
or @605	Instruction	0.1126	-0.8379	0.0242
phrase @606	Instruction	0.1197	-0.8292	0.0249
. @607	Instruction	0.1257	-0.8216	0.0242
ASSISTANT @608,609,610,611	Assistant/template	0.1690	-0.7745	0.0855

Template [space] immediately after Image is distinct from the ASSISTANT word. Negative Excess and Enrichment below1 mean the observed mass is below the uniform per-key baseline.

15 | 03539455 | ChooseRel | rescue

NATIVE FULL OBSERVATION | layer heatmap and omitted Top5 | no new answer generation



S: historical Semantic-selected; * omitted. Orange labels: omitted Question top5 in L0-15. P-1 excluded. L31 is observational only.

Omitted whole-group Top5 by scope (all32 heads averaged, no single-head maximum ranking):

L8-11

Question: the @584 0.138; Is @583 0.115; the @594 0.110; the @587 0.103; the @591 0.078

Instruction: word @604 0.065; Answer @598 0.054; . @607 0.050; the @599 0.047; question @600 0.041

Assistant/template: [space] @581 0.314; [NL] @582 0.192; ASSISTANT @608 0.106; [NL] @597 0.071

L0-15

Question: the @584 0.241; Is @583 0.216; the @587 0.211; the @594 0.191; the @591 0.176

Instruction: the @599 0.137; Answer @598 0.131; word @604 0.129; question @600 0.128; . @607 0.126

Assistant/template: [space] @581 0.334; [NL] @582 0.250; ASSISTANT @608 0.169; [NL] @597 0.149

L16-19

Question: the @584 0.244; the @594 0.207; Is @583 0.164; the @587 0.142; the @591 0.072

Instruction: Answer @598 0.057; the @599 0.045; question @600 0.043; or @605 0.041; using @601 0.038

Assistant/template: [space] @581 0.192; ASSISTANT @608 0.132; [NL] @582 0.091; [NL] @597 0.052

L31 R outputs cannot propagate onward to the first-answer P-1 position in the answer-image-blocked causal graph. High L31 mass is not evidence of functional necessity. All answers shown are historical, not Native Full generation in this run.

On which side of the picture is the man?



Gold: left
Historical All-Writer: Right
Historical Semantic-k: Left

P-1 is separate (not in R ranking):
L8-11: M=0.0686
L0-15: M=0.1529
L16-19: M=0.0865
L0-31: M=0.1106

Question groups below: L0-15 averages over all heads/layers/subwords. M=Mass, E=Excess, En=mean per-head Enrichment. Green=historical selected; gray=omitted; orange border=relative omitted top5, not a causal threshold.

On	which	side	of	the	picture
@583 n=1 M 0.2252 E -0.7611 En 0.0272	@584 n=1 M 0.2040 E -0.7807 En 0.0286	@585 n=1 M 0.1941 E -0.7889 En 0.0304	@586 n=1 M 0.1804 E -0.8009 En 0.0266	@587 n=1 M 0.2065 E -0.7731 En 0.0265	@588 n=1 M 0.1633 E -0.8146 En 0.0273

is	the	man	?
@589 n=1 M 0.1449 E -0.8314 En 0.0235	@590 n=1 M 0.1833 E -0.7913 En 0.0253	@591 n=1 M 0.2381 E -0.7348 En 0.0425	@592 n=1 M 0.1526 E -0.8188 En 0.0320

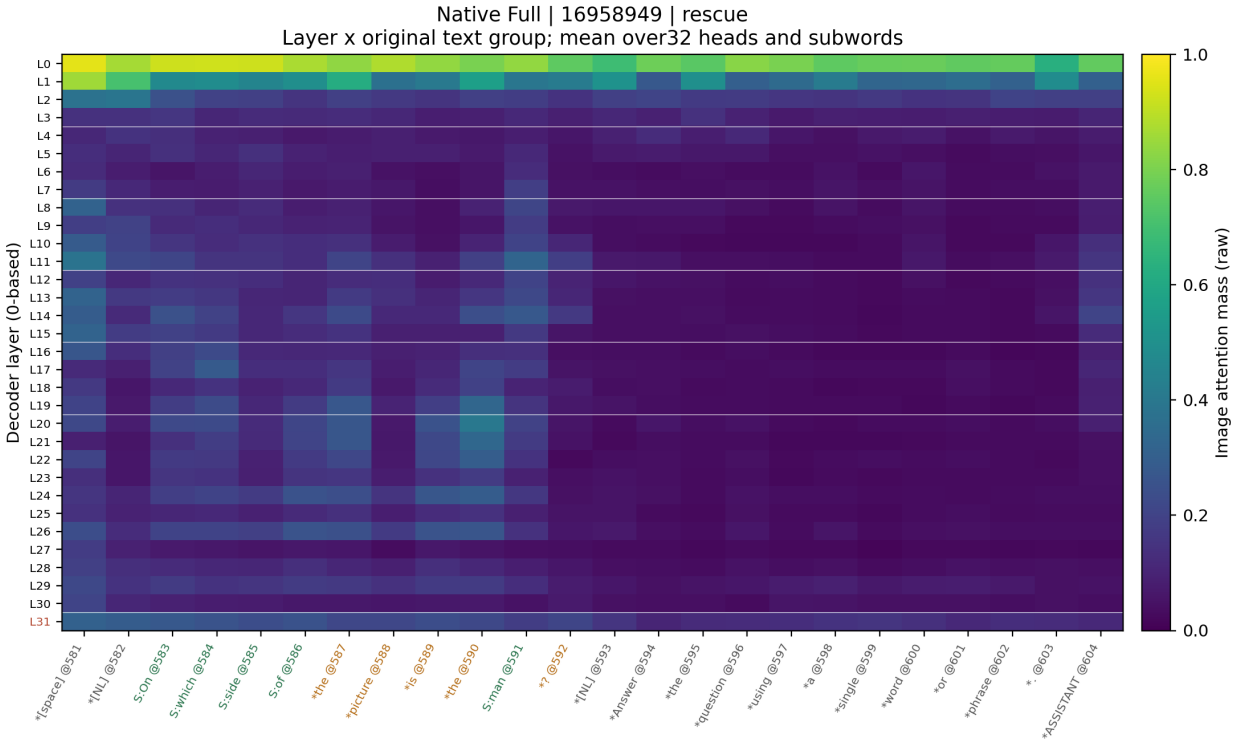
Other R groups (Instruction and Assistant/template); all historically omitted. Exact subwords and all-layer metrics are preserved in CSV.

Group / positions	Scope	Mass	Excess	Enrich.
[space] @581	Assistant/template	0.3231	-0.6666	0.0551
[NL] @582	Assistant/template	0.2427	-0.7453	0.0255
[NL] @593	Assistant/template	0.1334	-0.8363	0.0210
Answer @594	Instruction	0.1245	-0.8436	0.0199
the @595	Instruction	0.1322	-0.8342	0.0233
question @596	Instruction	0.1204	-0.8444	0.0256
using @597	Instruction	0.1129	-0.8503	0.0245
a @598	Instruction	0.1164	-0.8452	0.0234
single @599	Instruction	0.1104	-0.8496	0.0236
word @600	Instruction	0.1199	-0.8385	0.0242
or @601	Instruction	0.1082	-0.8486	0.0231
phrase @602	Instruction	0.1108	-0.8444	0.0236
. @603	Instruction	0.1206	-0.8331	0.0231
ASSISTANT @604,605,606,607	Assistant/template	0.1699	-0.7798	0.0561

Template [space] immediately after Image is distinct from the ASSISTANT word. Negative Excess and Enrichment below1 mean the observed mass is below the uniform per-key baseline.

16 | 16958949 | QueryAttr | rescue

NATIVE FULL OBSERVATION | layer heatmap and omitted Top5 | no new answer generation



S: historical Semantic-selected; * omitted. Orange labels: omitted Question top5 in L0-15. P-1 excluded. L31 is observational only.

Omitted whole-group Top5 by scope (all32 heads averaged, no single-head maximum ranking):

L8-11

Question: the @587 0.134; the @590 0.111; ? @592 0.102; picture @588 0.082; is @589 0.054

Instruction: word @600 0.056; . @603 0.050; Answer @594 0.050; the @595 0.042; a @598 0.035

Assistant/template: [space] @581 0.292; [NL] @582 0.190; ASSISTANT @604 0.113; [NL] @593 0.051

L0-15

Question: the @587 0.206; the @590 0.183; picture @588 0.163; ? @592 0.153; is @589 0.145

Instruction: the @595 0.132; Answer @594 0.124; . @603 0.121; question @596 0.120; word @600 0.120

Assistant/template: [space] @581 0.323; [NL] @582 0.243; ASSISTANT @604 0.170; [NL] @593 0.133

L16-19

Question: the @590 0.215; the @587 0.179; is @589 0.128; picture @588 0.084; ? @592 0.059

Instruction: Answer @594 0.039; or @601 0.036; question @596 0.036; the @595 0.033; using @597 0.030

Assistant/template: [space] @581 0.188; ASSISTANT @604 0.095; [NL] @582 0.087; [NL] @593 0.043

L31 R outputs cannot propagate onward to the first-answer P-1 position in the answer-image-blocked causal graph. High L31 mass is not evidence of functional necessity. All answers shown are historical, not Native Full generation in this run.